

SURVEY PAPER

Recent Advances in Gradient Algorithms for Optimal Control Problems^{1,2}

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Dedicated to Professor A. Busemann

Abstract. This paper summarizes recent advances in the area of gradient algorithms for optimal control problems, with particular emphasis on the work performed by the staff of the Aero-Astronautics Group of Rice University.

The following basic problem is considered: minimize a functional I which depends on the state $x(t)$, the control $u(t)$, and the parameter π . Here, I is a scalar, x an n -vector, u an m -vector, and π a p -vector. At the initial point, the state is prescribed. At the final point, the state x and the parameter π are required to satisfy q scalar relations. Along the interval of integration, the state, the control, and the parameter are required to satisfy n scalar differential equations.

First, the sequential gradient-restoration algorithm and the combined gradient-restoration algorithm are presented. The descent properties of these algorithms are studied, and schemes to determine the optimum stepsize are discussed. Both of the above algorithms require the solution of a linear, two-point boundary-value problem at each iteration. Hence, a discussion of integration techniques is given.

Next, a family of gradient-restoration algorithms is introduced. Not only does this family include the previous two algorithms as particular cases, but it allows one to generate several additional

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algorithms, namely, those with alternate restoration and optional restoration.

Then, two modifications of the sequential gradient-restoration algorithm are presented in an effort to accelerate terminal convergence. In the first modification, the quadratic constraint imposed on the variations of the control is modified by the inclusion of a positive-definite weighting matrix (the matrix of the second derivatives of the Hamiltonian with respect to the control). The second modification is a conjugate-gradient extension of the sequential gradient-restoration algorithm.

Next, the addition of a nondifferential constraint, to be satisfied everywhere along the interval of integration, is considered. In theory, this seems to be only a minor modification of the basic problem. In practice, the change is considerable in that it enlarges dramatically the number and variety of problems of optimal control which can be treated by gradient-restoration algorithms. Indeed, by suitable transformations, almost every known problem of optimal control theory can be brought into this scheme. This statement applies, for instance, to the following situations: (i) problems with control equality constraints, (ii) problems with state equality constraints, (iii) problems with equality constraints on the time rate of change of the state, (iv) problems with control inequality constraints, (v) problems with state inequality constraints, and (vi) problems with inequality constraints on the time rate of change of the state.

Finally, the simultaneous presence of nondifferential constraints and multiple subarcs is considered. The possibility that the analytical form of the functions under consideration might change from one subarc to another is taken into account. The resulting formulation is particularly relevant to those problems of optimal control involving bounds on the control or the state or the time derivative of the state. For these problems, one might be unwilling to accept the simplistic view of a continuous extremal arc. Indeed, one might want to take the more realistic view of an extremal arc composed of several subarcs, some internal to the boundary being considered and some lying on the boundary.

The paper ends with a section dealing with transformation techniques. This section illustrates several analytical devices by means of which a great number of problems of optimal control can be reduced to one of the formulations presented here. In particular, the following topics are treated: (i) time normalization, (ii) free initial state, (iii) bounded control, and (iv) bounded state.

Key Words. Survey papers, gradient methods, numerical methods, computing methods, calculus of variations, optimal control, gradient-restoration algorithms, boundary-value problems, bounded control problems, bounded state problems, nondifferential constraints.

1. Introduction

In every branch of science and engineering, there exist systems which are controllable, that is, they can be made to behave in different ways depending on the will of the operator. Every time the operator of a system exercises an option, a choice in the distribution of the quantities controlling the system, he produces a change in the distribution of the states occupied by the system and, hence, a change in the final state. Therefore, it is natural to pose the following question: Among all the admissible options, what is the particular option which renders the system optimum? As an example, what is the option which minimizes the difference between the final value and the initial value of an arbitrarily specified function of the state of the system? The body of knowledge covering problems of this type is called optimal control theory. Applications occur in every field of science and engineering, and also in economics.

For example, consider one field of problems, that of aerospace engineering. The pilot of a jet-propelled aircraft climbing from take-off to cruise can change the flight time by controlling the time history of the elevator position and the thrust setting; hence, a minimum-time problem arises. The astronaut of a spaceship being transferred from one orbit to another can change the amount of propellant consumed by simultaneously controlling the time history of the thrust modulus and the thrust direction; hence, it is natural to seek the minimum propellant consumption. The ballistician who designs the ogive of an artillery projectile can change the aerodynamic drag by controlling the distribution of longitudinal slopes versus the abscissa; hence, a minimum-drag problem arises. Finally, the aerodynamicist who designs a rocket nozzle can change the thrust by controlling the distribution of longitudinal slopes along the abscissa; hence, it is natural to seek the maximum thrust.

Of course, applications of optimal control theory are by no means restricted to aerospace engineering. A typical problem of chemical engineering is the determination of the optimal catalyst blending with successive reactions in a tubular reactor or the optimal start-up procedure for a chemical reactor. A typical problem of civil engineering is the design of the beam having minimum weight for a given loading or the

design of the axisymmetric dome having minimum weight for given loading. A typical problem of economics is the optimal planning for inventory and production or the study of the optimal economic growth of a system.

Note that only a minority of current engineering problems can be solved by purely analytical methods (Refs. 94–107). Hence, it is important to develop numerical techniques enabling one to solve optimal control problems on a digital computer. These numerical techniques can be classified into two groups: first-order methods and second-order methods. First-order methods (or gradient methods) are those techniques which employ at most the first derivatives of the functions under consideration (Refs. 12–43, Refs. 60–77, and Refs. 108–132). Second-order methods (or quasilinearization methods) are those techniques which employ at most the second derivatives of the functions under consideration (Refs. 44–48, Refs. 78–81, and Refs. 133–147). Both gradient methods and quasilinearization methods require the solution of a linear, two-point boundary-value problem (and sometimes the solution of a linear, multi-point boundary-value problem) at every iteration. This being the case, progress in the area of numerical methods for differential equations (Refs. 1–11, Refs. 49–59, and Refs. 82–93) is essential to the efficient solution of optimal control problems on a digital computer.

In this paper, we review recent advances in the area of gradient methods for optimal control problems. Because of space limitations, we make no attempt to cover every possible technique and every possible approach, a material impossibility in view of the large number of publications available. Thus, except for noting the early work performed by Kelley (Refs. 110 and 113) and Bryson (Refs. 111–112 and 114–115), we devote the body of the paper to a review of the work performed in recent years by the Aero-Astronautics Group of Rice University (Refs. 1–81).

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1.1. Notation. In this paper, vector–matrix notation is used for conciseness. A vector, such as x , is defined to be a column vector. A matrix, such as φ_u , is defined so that its dimensions are $\dim(u) \times \dim(\varphi)$. A three-dimensional array, such as $\varphi_{u\pi}$, is defined so that its dimensions are $\dim(\pi) \times \dim(u) \times \dim(\varphi)$.

The dot sign denotes derivative with respect to the time t , that is,

$\dot{x} = dx/dt$. The symbol T denotes transposition of vector or matrix.

For a single-subarc problem, the subscript 0 denotes the initial point and the subscript 1 denotes the final point. The time t is normalized, so that $t = 0$ denotes the initial point and $t = 1$ denotes the final point.

For a double-subarc problem, the subscripts 0, 1, 2 denote the initial point, the endpoint of the first subarc, and the final point. The time t is normalized, so that $t = 0$ denotes the initial point, $t = 1$ denotes the endpoint of the first subarc, and $t = 2$ denotes the final point.

For a triple-subarc problem, the subscripts 0, 1, 2, 3 denote the initial point, the endpoint of the first subarc, the endpoint of the second subarc, and the final point. The time t is normalized, so that $t = 0$ denotes the initial point, $t = 1$ denotes the endpoint of the first subarc, $t = 2$ denotes the endpoint of the second subarc, and $t = 3$ denotes the final point.

In a multiple subarc problem, the subscript $-$ denotes conditions preceding a corner point, and the subscript $+$ denotes conditions following a corner point.

2. Basic Problem

We initiate our review by considering the following basic problem.

Problem (A-1). Minimize the functional

$$I = \int_0^1 f(x, u, \pi, t) dt + [g(x, \pi, t)]_1, \quad (1)$$

with respect to the state $x(t)$, the control $u(t)$, and the parameter π which satisfy the constraints

$$\dot{x} - \varphi(x, u, \pi, t) = 0, \quad 0 \leq t \leq 1, \quad (2)$$

$$(x)_0 = \text{given}, \quad (3)$$

$$[\psi(x, \pi, t)]_1 = 0. \quad (4)$$

In Eqs. (1)–(4), the quantities I, f, g, t are scalar and the quantities x, u, π, φ, ψ are vectors having dimensions n, m, p, n, q , respectively.

2.1. Augmented Functional. The previous Problem (A-1) is one of the Bolza type and is characterized by the augmented functional (Refs. 102-104)

$$J = \int_0^1 [f + \lambda^T(\dot{x} - \varphi)] dt + (g + \mu^T\psi)_1, \quad (5)$$

where λ and μ are vector Lagrange multipliers having dimensions n and q , respectively. It is known that the minimization of (1), subject to (2)-(4), is identical with the minimization of (5), subject to (2)-(4), regardless of the choice of the multipliers $\lambda(t)$ and μ .

2.2. First-Order Conditions. If the multipliers $\lambda(t)$ and μ are chosen consistently with

$$\dot{\lambda} - f_x + \varphi_x \lambda = 0, \quad 0 \leq t \leq 1, \quad (6)$$

$$(\lambda + g_x + \psi_x \mu)_1 = 0, \quad (7)$$

then the optimum solution $x(t)$, $u(t)$, π must satisfy the following relations:

$$f_u - \varphi_u \lambda = 0, \quad 0 \leq t \leq 1, \quad (8)$$

$$\int_0^1 (f_\pi - \varphi_\pi \lambda) dt + (g_\pi + \psi_\pi \mu)_1 = 0, \quad (9)$$

which, together with (6)-(7), constitute the *optimality conditions*.

2.3. Two-Point Boundary-Value Problem. The system composed of the feasibility equations (2)-(4) and the optimality conditions (6)-(9) constitutes a nonlinear, two-point boundary-value problem in which the unknowns are the functions $x(t)$, $u(t)$, π and the multipliers $\lambda(t)$, μ . In scalar terminology, this problem involves $2n + m$ unknown functions [the components of the vectors $x(t)$, $\lambda(t)$, $u(t)$] and $p + q$ unknown parameters (the components of the vectors π and μ). These unknown functions and parameters must be determined so as to satisfy the $2n$ differential equations (2) and (6), the m nondifferential equations (8), the p integral conditions (9), the n initial conditions (3), and the $q + n$ final conditions (4) and (7). Closed-form solutions are possible only for particular cases. In general, numerical methods must be employed.

2.4. Linear-Quadratic Case. If the functions f and g appearing in the functional (1) are quadratic in x , u , π , and if the functions φ and ψ appearing in the constraints (2)-(4) are linear in x , u , π , the system

composed of the feasibility equations (2)–(4) and the optimality conditions (6)–(9) is linear in $x(t)$, $u(t)$, π , $\lambda(t)$, μ . Because of the linearity, several general techniques can be employed in order to arrive at the numerical solution in a single iteration. These techniques are:

- (i) the method of complementary functions (Ref. 83),
- (ii) the method of the adjoint variables (Ref. 83),
- (iii) the method of particular solutions (Ref. 49).

If the method of particular solutions is employed, the total number of integrations which must be performed in this single iteration is $n + p + 1$, where n is the dimension of the state vector x and p is the dimension of the parameter π .

2.5. General Case. If the functions f and g are nonquadratic in x , u , π , and/or if the functions φ and ψ are nonlinear in x , u , π , the system composed of the feasibility equations (2)–(4) and the optimality conditions (6)–(9) is nonlinear in $x(t)$, $u(t)$, π , $\lambda(t)$, μ . Under these conditions, iterative methods must be employed in order to arrive at some approximation to the true solution.

Depending on whether these numerical methods employ at most the first derivatives or at most the second derivatives of the functions under consideration, two classes of algorithms can be developed: first-order algorithms, also called gradient methods (see Refs. 12–43, Refs. 60–77, and Refs. 108–132) and second-order algorithms, also called quasilinearization methods (see Refs. 44–48, Refs. 78–81, and Refs. 133–147).

2.6. Performance Indexes. When solving a problem numerically on a digital computer, it is necessary to define convergence in a numerical sense. In this connection, let the norm of a vector y be defined as

$$N(y) = y^T y. \quad (10)$$

Then, let P and Q denote the scalar performance indexes

$$P = \int_0^1 N(\dot{x} - \varphi) dt + N(\psi)_1, \quad (11)$$

$$Q = \int_0^1 N(\dot{\lambda} - f_x + \varphi_x \lambda) dt + \int_0^1 N(f_u - \varphi_u \lambda) dt \\ + N \left[\int_0^1 (f_\pi - \varphi_\pi \lambda) dt + (g_\pi + \psi_\pi \mu)_1 \right] + N(\lambda + g_x + \psi_x \mu)_1, \quad (12)$$

which measure the errors in the constraints and the optimality conditions, respectively. Finally, let R denote the sum

$$R = P + Q, \quad (13)$$

which is the cumulative error in the system.

Observe that

$$P = 0, \quad Q = 0, \quad R = 0, \quad (14)$$

for the exact variational solution and that

$$P > 0 \quad \text{and/or} \quad Q > 0, \quad \text{so that} \quad R > 0, \quad (15)$$

for any approximation to the solution. This being the case, numerical convergence can be defined as follows: an iterative algorithm will be stopped whenever functions $x(t)$, $u(t)$, π and multipliers $\lambda(t)$, μ are found such that

$$P \leq \epsilon_1, \quad Q \leq \epsilon_2, \quad (16)$$

where ϵ_1 and ϵ_2 are small, preselected numbers. Alternatively, (16) can be replaced by

$$R \leq \epsilon_3, \quad (17)$$

where ϵ_3 is another small, preselected number.

Use of Ineqs. (16) should be preferred for a first-order method, and use of Ineq. (17) should be preferred for a second-order method. This is because P and Q do not decrease at the same rate in a first-order method, while they do decrease at the same rate in a second-order method.

With particular reference to a first-order method, the tolerance parameters ϵ_1 and ϵ_2 need not be the same. Indeed, they can be chosen so that

$$\epsilon_1 \ll \epsilon_2. \quad (18)$$

This is because, in the real world, constraint satisfaction is generally more important than satisfaction of the optimality conditions (an obvious illustration is the rendezvous between two spaceships in orbit, as illustrated by the recent Apollo-Soyuz flight and by the forthcoming space shuttle flights). In addition, the nature of engineering optima is rather flat: nothing catastrophic generally happens (from the point of view of the value of the functionals I and/or J) if one chooses to satisfy the optimality conditions (6)–(9) with a relaxed accuracy, rather than in a more exact way.

With the above considerations in mind, the following are examples of stopping conditions which can be employed in a first-order method (Refs. 12-43):

$$P \leq 10^{-8}, \quad Q \leq 10^{-4}, \quad (19)$$

and in a second-order method (Refs. 44-48):

$$R \leq 10^{-12}. \quad (20)$$

Of course, implicit in the use of (19) or (20) is the assumption that the quantities appearing in the optimal control problem have been normalized, so that the expected solutions are characterized by quantities of order one.

3. Sequential Gradient-Restoration Algorithm

3.1. Preliminary Considerations. The sequential gradient-restoration algorithm is an iterative technique which includes a sequence of two-phase processes or cycles, each composed of a gradient phase and a restoration phase. This technique is constructed so that the following properties are satisfied at the beginning and at the end of each cycle (Ref. 63).

Property (i). The functions $x(t)$, $u(t)$, π available both at the beginning and at the end of each cycle must be feasible; that is, they must be consistent with the constraints (2)-(4) within the preselected accuracy (16-1).

Property (ii). The functions $x(t)$, $u(t)$, π produced at the end of each cycle must be characterized by a value of the functional I [see Eq. (1)] which is smaller than that associated with the functions available at the beginning of the cycle.

Because of (i) and (ii), it is clear that the following additional property must be satisfied.

Property (iii). The functions $x(t)$, $u(t)$, π produced at the end of each cycle must be characterized by a value of the augmented functional J [see Eq. (5)] which is smaller than that associated with the functions available at the beginning of the cycle.

To achieve the above properties, each cycle is made of two phases, a gradient phase and a restoration phase.

The *gradient phase* is started only when Ineq. (16-1) is satisfied and involves a single iteration. In this single iteration, the objective is to

reduce the functional I or the augmented functional J , while the constraints are satisfied to first order and the norm of the variations of the control and the parameter is held constant.

The *restoration phase* is started only when Ineq. (16-1) is violated and involves one or more iterations. In each restorative iteration, the objective is to reduce the functional P , while the constraints are satisfied to first order and the norm of the variations of the control and the parameter is minimized. The restoration phase is terminated whenever Ineq. (16-1) is satisfied.

During each iteration, gradient or restorative, use of nonlinear equations must be avoided. Therefore, the exact feasibility equations (2)–(4) are replaced by their corresponding linearized approximations. These linearized approximations do not include forcing terms in the gradient phase, while they do include forcing terms in the restoration phase.

3.2. Notation. For any iteration of the gradient phase or the restoration phase, the following terminology is adopted: $x(t)$, $u(t)$, π denote the nominal functions; $\tilde{x}(t)$, $\tilde{u}(t)$, $\tilde{\pi}$ denote the varied functions; and $\Delta x(t)$, $\Delta u(t)$, $\Delta \pi$ denote the displacements leading from the nominal functions to the varied functions. These quantities satisfy the definitions

$$\tilde{x}(t) = x(t) + \Delta x(t), \quad \tilde{u}(t) = u(t) + \Delta u(t), \quad \tilde{\pi} = \pi + \Delta \pi. \quad (21)$$

If the variations appearing in (21) are linear in the stepsize α , where $\alpha > 0$, they take the form

$$\Delta x(t) = \alpha A(t), \quad \Delta u(t) = \alpha B(t), \quad \Delta \pi = \alpha C, \quad (22)$$

with the implication that

$$\tilde{x}(t) = x(t) + \alpha A(t), \quad \tilde{u}(t) = u(t) + \alpha B(t), \quad \tilde{\pi} = \pi + \alpha C. \quad (23)$$

The functions $A(t)$, $B(t)$, C must be determined so as to produce some desirable effect at every iteration, namely, the decrease of the functionals J and/or P . Thus, the following descent properties are required:

$$\tilde{J} < J \quad \text{and/or} \quad \tilde{P} < P, \quad (24)$$

where J , P are associated with the nominal functions and $\tilde{J}$, $\tilde{P}$ are associated with the varied functions. Inequality (24-1) characterizes the gradient phase, and Ineq. (24-2) characterizes the restoration phase.

In turn, relations (24) can be enforced at every iteration providing

the stepsize α is sufficiently small and the functions $A(t)$, $B(t)$, C are chosen so that

$$\delta J < 0 \quad \text{and/or} \quad \delta P < 0, \quad (25)$$

where the symbol $\delta(\dots)$ denotes the first variation. Inequality (25-1) characterizes the gradient phase, and Ineq. (25-2) characterizes the restoration phase.

Clearly, then, every iteration includes two basic operations: (a) the determination of functions $A(t)$, $B(t)$, C consistent with the first variation requirements (25); and (b) the determination of the stepsize α consistent with the total variation requirements (24).

3.3. Determination of the Basic Functions. There exist an infinite number of combinations of functions $A(t)$, $B(t)$, C capable of satisfying the first-variation inequalities (25), subject to the linearized constraints. In order to arrive at a unique combination of functions, some additional requirement must be imposed. This is done through the formulation of the following auxiliary minimization problems.

Problem (A-2). For the gradient phase, minimize the functional

$$\delta I = \int_0^1 \delta f \, dt + (\delta g)_1, \quad (26)$$

with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (27)$$

$$\delta(x)_0 = 0, \quad (28)$$

$$\delta(\psi)_1 = 0, \quad (29)$$

and the isoperimetric constraint

$$K = \int_0^1 \Delta u^T \Delta u \, dt + \Delta \pi^T \Delta \pi. \quad (30)$$

Note the absence of forcing terms from Eqs. (27)–(29). This implies that the nominal functions characterizing the gradient phase satisfy the constraints (2)–(4) within the preselected degree of accuracy (16-1). Also note that the isoperimetric constraint (30) is imposed in order to prevent the variations $\Delta x(t)$, $\Delta u(t)$, $\Delta \pi$ from being infinitely large. However, the actual level of the isoperimetric constant K need not be specified *a priori* (see Remark 4.7).

Problem (A-3). For the restoration phase, minimize the functional (30), with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) + \alpha(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (31)$$

$$\delta(x)_0 = 0, \quad (32)$$

$$\delta(\psi)_1 + \alpha(\psi)_1 = 0, \quad (33)$$

where α is a prescribed quantity in the range $0 \leq \alpha \leq 1$.

Note that forcing terms are absent from Eq. (32), but are present in Eqs. (31) and (33). This implies that the nominal functions characterizing the restoration phase satisfy the initial condition (3), but might violate the differential equation (2) and/or the final condition (4). Indeed, it is the purpose of the restoration phase to correct these violations, while causing the least possible disturbance in the system (Ref. 61). This is the significance of the least-square criterion (30).

3.4. Linear, Two-Point Boundary-Value Problem. Problems (A-2) and (A-3) are variational problems of the Bolza type, each governed by a different set of linearized constraints and optimality conditions. However, the form of these linearized constraints and optimality conditions is such that the equations governing Problems (A-2) and (A-3) can be embedded in a single system. For compactness, as well as to facilitate programming, this point of view is taken here, and the single system governing both the gradient phase and the restoration phase is written as follows:

$$\dot{A} - \varphi_x^T A - \varphi_u^T B - \varphi_\pi^T C + k_r(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (34)$$

$$(A)_0 = 0, \quad (35)$$

$$(\psi_x^T A + \psi_\pi^T C + k_r \psi)_1 = 0, \quad (36)$$

and

$$\dot{\lambda} - k_g f_x + \varphi_x \lambda = 0, \quad 0 \leq t \leq 1, \quad (37)$$

$$(\lambda + k_g g_x + \psi_x \mu)_1 = 0, \quad (38)$$

$$B + k_g f_u - \varphi_u \lambda = 0, \quad 0 \leq t \leq 1, \quad (39)$$

$$C + \int_0^1 (k_g f_\pi - \varphi_\pi \lambda) dt + (k_g g_\pi + \psi_\pi \mu)_1 = 0. \quad (40)$$

The constants k_g and k_r appearing in (34)–(40) take the following values:

$$\text{gradient phase, } k_g = 1, \quad k_r = 0; \quad (41)$$

$$\text{restoration phase, } k_g = 0, \quad k_r = 1. \quad (42)$$

For given nominal functions $x(t)$, $u(t)$, π and constants k_g and k_r , Eqs. (34)–(40) define the functions $A(t)$, $B(t)$, C and the multipliers $\lambda(t)$ and μ . Clearly, Eqs. (34)–(40) constitute a linear, two-point boundary-value problem, which can be solved independently of the value assigned to the stepsize α .

3.5. Method of Particular Solutions. Let a *sweep* be defined as a backward-forward integration of Eqs. (34)–(40), obtained by specifying the q components of the multiplier μ and satisfying all of the equations, except (36). For given values of the constants k_g and k_r , a sweep is described by the following procedure: (a) assign a value to the multiplier μ ; (b) determine $\lambda(1)$ with Eq. (38); (c) compute $\lambda(t)$ by backward integration of Eq. (37), subject to the final condition (38); (d) compute $B(t)$ with (39); (e) compute C with (40); and (f) compute $A(t)$ by forward integration of (34), subject to the initial condition (35).

If the multiplier μ is chosen arbitrarily, a single sweep does not yield values of $A(1)$ and C consistent with the final condition (36). However, since both μ and ψ are q -vectors, one can find in principle the special value of μ which leads to the satisfaction of the final condition (36).

If the method of particular solutions is employed, one must execute $q + 1$ independent sweeps, each obtained through a different specification of the components of the multiplier μ . Specifically, the first q sweeps are performed by considering the identity matrix of order q and by choosing the vector μ associated with the i th integration, $i = 1, \dots, q$, as identical with the i th column of the above identity matrix. The last sweep is performed by choosing the vector μ associated with the $(q + 1)$ th integration as identical with the null vector.

Upon combining the particular solutions linearly, and upon writing the composite solution in the form

$$A(t) = \sum k_i A_i(t), \quad B(t) = \sum k_i B_i(t), \quad C = \sum k_i C_i, \quad (43)$$

$$\lambda(t) = \sum k_i \lambda_i(t), \quad \mu = \sum k_i \mu_i, \quad (44)$$

where $i = 1, \dots, q + 1$, the $q + 1$ constants of the combination are determined by satisfying the q final conditions (36) and the supplementary scalar relation $\sum k_i = 1$, which arises from the nonhomogeneity of the system (34)–(40).

3.6. Computational Considerations. Each sweep involves integrating $2n$ differential equations, that is, the n multiplier equations (37) and the n linearized state equations (34). Since $q + 1$ sweeps are

involved, the amount of computational work performed per iteration, gradient or restorative, is proportional to the factor⁴

$$w = 2n(q + 1). \quad (45)$$

If the parameter π is absent, the number of final conditions q is such that $0 \leq q \leq n$; hence, the smallest value of the factor (45) is

$$w = 2n \quad (46)$$

for a free endpoint problem ($q = 0$). Conversely, the largest value of the factor (45) is

$$w = 2n(n + 1) \quad (47)$$

for a fixed endpoint problem ($q = n$). Thus, for a free endpoint problem, the computational effort per iteration increases linearly with the dimension of the state vector. On the other hand, for a fixed endpoint problem, the computational effort per iteration increases with the square of the dimension of the state vector.

3.7. Descent Properties. The functions $A(t)$, $B(t)$, C solving Eqs. (34)–(40) are such that the following first-variation properties, consistent with the requirements (25), hold:⁵

$$\text{gradient phase, } \delta J = -\alpha Q; \quad (48)$$

$$\text{restoration phase, } \delta P = -2\alpha P. \quad (49)$$

Because of these first-variation properties, it is possible in principle to determine the gradient stepsize so that the descent property (24-1) is enforced in a gradient iteration; it is also possible to determine the restoration stepsize so that the descent property (24-2) is enforced in a restorative iteration.

3.8. Determination of the Stepsize. Suppose now that the functions $A(t)$, $B(t)$, C are known by solving the linear, two-point boundary-value problem (34)–(40). Since the nominal functions $x(t)$, $u(t)$, π are also known, the varied functions constitute the following one-parameter family of solutions:

$$\tilde{x}(t) = x(t) + \alpha A(t), \quad \tilde{u}(t) = u(t) + \alpha B(t), \quad \tilde{\pi} = \pi + \alpha C. \quad (50)$$

⁴ For problems involving a parameter π , the permissible range of final conditions is $0 \leq q \leq n + p$.

⁵ Note that Q is given by Eq. (70).

For this one-parameter family, the augmented functional (5) and the constraint error (11) take the form

$$\tilde{J} = \tilde{J}(\alpha), \quad \tilde{P} = \tilde{P}(\alpha). \quad (51)$$

The stepsize α must be determined so that the following inequalities are enforced at every iteration:

$$\text{gradient phase, } \tilde{J}(\alpha) < \tilde{J}(0), \quad \tilde{P}(\alpha) \leq P_*; \quad (52)$$

$$\text{restoration phase, } \tilde{P}(\alpha) < \tilde{P}(0). \quad (53)$$

In (52), the symbol P_* denotes a preselected upper bound (not necessarily a small number) for the constraint violation which is permissible during the gradient phase. While more detailed considerations on the determination of the stepsize are given in Sections 3.9 and 3.10, the following remarks can be made at this point.

(i) In the search for the gradient stepsize, use of the functional $\tilde{J}$ is to be preferred to use of the functional I . This is clear if one considers an obvious limiting case: that of a linear functional (1), subject to nonlinear constraints (2)–(4). In this case, the function $\tilde{J}(\alpha)$ is monotonically decreasing with respect to α , while the function $\tilde{J}(\alpha)$ might exhibit a natural minimum with respect to α for nominal functions $x(t)$, $u(t)$, π sufficiently close to the solution.

(ii) In the search for the gradient stepsize, Ineq. (52-2) plays a two-fold role. First, it limits the amount of constraint violation which is permissible during the gradient phase; this facilitates the subsequent restoration phase. Second, it guarantees the existence of a finite value for the stepsize in those cases where the function $\tilde{J}(\alpha)$ is monotonically decreasing with respect to α . For certain problems, this situation might arise in the beginning stages of a sequential gradient–restoration algorithm if the nominal functions $x(t)$, $u(t)$, π are not sufficiently close to the solution.

3.9. Gradient Stepsize. There exist various procedures (search techniques) which guarantee the enforcement of Ineqs. (52) during a gradient iteration, for instance,

- (i) single-step quadratic interpolation,
- (ii) multi-step quadratic interpolation,
- (iii) single-step cubic interpolation,
- (iv) multi-step cubic interpolation.

With reference to (i)–(iv), the terminology is as follows. Quadratic interpolation denotes a technique by which $J(\alpha)$ is approximated with a quadratic function over some interval of values of α ; analogously, cubic interpolation denotes a technique by which $J(\alpha)$ is approximated with a cubic function. Also, a single-step procedure denotes the non-iterative use of quadratic interpolation or cubic interpolation; analogously, a multi-step procedure denotes the iterative use of quadratic interpolation or cubic interpolation.

Basic to the above search techniques is the consideration of a sequence of stepsizes, with elements in an order established *a priori*. By scanning the values of $J(\alpha)$ and/or $\tilde{J}_\alpha(\alpha)$ for the elements of the sequence, one can delimit an interval $\alpha_1 \leq \alpha \leq \alpha_2$ containing the minimum point of $J(\alpha)$. Once this interval is found, the augmented functional $\tilde{J}(\alpha)$ can be represented in the following polynomial fashion:

$$\text{quadratic interpolation, } \tilde{J}(\alpha) = C_0 + C_1\alpha + C_2\alpha^2; \quad (54)$$

$$\text{cubic interpolation, } \tilde{J}(\alpha) = C_0 + C_1\alpha + C_2\alpha^2 + C_3\alpha^3. \quad (55)$$

The coefficients of these polynomials are then determined by imposing a sufficient number of conditions (obviously, three for the quadratic interpolation scheme and four for the cubic interpolation scheme) on $\tilde{J}(\alpha)$ and/or $\tilde{J}_\alpha(\alpha)$ at α_1 , α_2 , and perhaps some intermediate stepsize α_3 in the range $\alpha_1 \leq \alpha_3 \leq \alpha_2$. With the coefficients known, the approximate optimal stepsize α_0 is computed by minimizing $\tilde{J}(\alpha)$ with respect to α . This leads to the following expressions:

$$\text{quadratic interpolation, } \alpha_0 = -C_1/2C_2; \quad (56)$$

$$\text{cubic interpolation, } \alpha_0 = (1/3C_3)[-C_2 + \sqrt{(C_2^2 - 3C_1C_3)}]. \quad (57)$$

In a single-step procedure, the approximate optimal stepsize α_0 is computed only once; this computation is followed by a bisection process, if needed, until satisfaction of Ineqs. (52) occurs. In a multi-step procedure, the technique described above is applied iteratively until one of the following stopping conditions is satisfied:

$$|\tilde{J}_\alpha(\alpha)| \leq \epsilon_4 \quad \text{or} \quad |\tilde{J}_\alpha(\alpha)/\tilde{J}_\alpha(0)| \leq \epsilon_5, \quad (58)$$

where ϵ_4 and ϵ_5 are small, preselected numbers.

It is of interest to examine the relative convenience of single-step procedures versus multi-step procedures. While the ordinary gradient method is not endowed with memory characteristics (as, for example, the conjugate-gradient method), it is a self-correcting method; specifically,

the control changes and parameter changes determined in a subsequent iteration automatically account for the fact that different varied functions are produced in a previous iteration because of the replacement of the exact optimal stepsize with some reasonable approximation. This being the case, one must expect the number of iterations for convergence with single-step procedures to be of the same order as that needed with multi-step procedures.

In conclusion, single-step search procedures are generally satisfactory for the ordinary gradient method. On the other hand, multi-step search procedures might become desirable, indeed essential, for the conjugate-gradient method (Refs. 72–73).

3.10. Restoration Stepsize. The computation of the restoration stepsize is much simpler than that of the gradient stepsize. This is because of the following property: if the constraints (2)–(4) are linear in x , u , π , the stepsize $\alpha = 1$ allows not only satisfaction of Ineq. (53), but indeed produces complete restoration in one iteration. In other words, for a linear system, the stepsize $\alpha = 1$ is such that $\tilde{P}(1) = 0$ at the end of one restorative iteration.

If the constraints (2)–(4) are nonlinear, the stepsize $\alpha = 1$ is not especially significant in the first few restorative iterations. But it becomes more and more significant toward the end of the restoration phase and as the sequential gradient–restoration algorithm progresses toward convergence. This is why there is no need, for the restoration phase, to resort to the search schemes (quadratic interpolation or cubic interpolation) of Section 3.9. It suffices to employ a simple bisection process, starting from $\alpha = 1$, until satisfaction of Ineq. (53) occurs.

3.11. Stability Considerations. Let I_1 denote the value of the functional (1) at the beginning of the gradient phase, let I_2 denote the value of (1) at the end of the gradient phase or beginning of the restoration phase, and let I_3 denote the value of (1) at the end of the restoration phase. Note that I_1 and I_2 are not comparable, since the constraints are not satisfied to the same accuracy. On the other hand, I_1 and I_3 are comparable, since the constraints are satisfied to the same accuracy.

To give stability to the sequential gradient–restoration algorithm, one must require satisfaction of the inequality

$$I_3 < I_1 \quad (59)$$

and proceed as follows. If Ineq. (59) is satisfied, the next gradient phase is started. If Ineq. (59) is violated, one returns to the previous gradient

phase and reduces the gradient stepsize (for example, one can use a bisection process) until, after restoration, Ineq. (59) is satisfied.

That the above procedure leads to satisfaction of Ineq. (59) is guaranteed by the fact that, if α_g is the gradient stepsize and α_r is the restoration stepsize, the gradient corrections and the restorative corrections have the following order of magnitude:

$$\text{gradient phase, } \Delta x(t), \Delta u(t), \Delta \pi \equiv O(\alpha_g); \quad (60)$$

$$\text{restoration phase, } \Delta x(t), \Delta u(t), \Delta \pi \equiv O(\alpha_r \alpha_g^2). \quad (61)$$

Hence, if the gradient stepsize is sufficiently small, one can enforce not only Ineq. (52-1), but also Ineq. (59). Thus, the restoration phase preserves the descent property of the gradient phase. This key property renders the sequential gradient-restoration algorithm computationally stable on a digital computer.

4. Remarks

In this section, we summarize several remarks pertinent to the development and implementation of sequential gradient-restoration algorithms on a digital computer.

Remark 4.1. For the gradient phase, Problem (A-2) could be modified by replacing the isoperimetric constraint (30) with the alternate expression

$$K = \int_0^1 (\Delta x^T \Delta x + \Delta u^T \Delta u) dt + \Delta \pi^T \Delta \pi. \quad (62)$$

An analogous remark holds for the restoration phase. Problem (A-3) could be modified by replacing the functional to be minimized (30) with the alternate expression (62).

In principle, it is possible to construct sequential gradient-restoration algorithms in this way. In practice, this is not desirable, since use of (62) in place of (30) would produce a coupling between the linearized state equation (34) and the modified form of the multiplier equation (37), that is,

$$\dot{\lambda} - A - k_g f_x + \varphi_x \lambda = 0, \quad 0 \leq t \leq 1. \quad (63)$$

As a consequence, the backward-forward integration procedure described in Section 3.5 would no longer be possible. The system (34)–(40), with

(37) replaced by (63), would have to be integrated either all in forward time or all in backward time. The number of sweeps required to solve the linear, two-point boundary-value problem would increase from $q + 1$ to $n + p + 1$, and the computational work per iteration would be proportional to the factor

$$w = 2n(n + p + 1), \quad (64)$$

regardless of the number of boundary conditions to be satisfied at the final point.

Remark 4.2. For the restoration phase, Problem (A-3) could further be modified by adding to the constraints (31)–(33) the supplementary isoperimetric constraint

$$\delta I = \int_0^1 \delta f \, dt + (\delta g)_1 = 0, \quad (65)$$

designed to avoid excessive change in the value of the functional (1).

In principle, it is possible to construct sequential gradient-restoration algorithms in this way. In practice, this modification of the restoration phase is not desirable for two reasons. First, use of (65) would cause an increase of the number of sweeps required to solve the linear, two-point boundary-value problem from $q + 1$ to $q + 2$; consequently, the computational work per restorative iteration would be proportional to the factor

$$w = 2n(q + 2). \quad (66)$$

In addition, the variations of the control and the parameter produced by solving this modification of Problem (A-3) would be larger than the variations of the control and the parameter produced by the solution of the original Problem (A-3). Furthermore, actual computational experience has shown that the inclusion of the isoperimetric constraint (65) might slow down the convergence of the sequential gradient-restoration algorithm. Occasionally, the inclusion of (65) might prevent convergence, by preventing constraint satisfaction.

In conclusion, this writer believes that the proper restoration mode is that described by the original Problem (A-3). Its solution yields the smallest changes of the control and the parameter compatible with constraint restoration. The associated state change cannot be too large, owing to the fact that the state change is related to the changes of the control and the parameter through the linearized form of the constraints (2)–(4).

Remark 4.3. The sequential gradient-restoration algorithm can be started with nominal functions $x(t)$, $u(t)$, π satisfying the initial condition (3) and violating the differential equation (2) and/or the final condition (4). If the constraint error P satisfies Ineq. (16-1), the algorithm starts with a gradient phase. On the other hand, if the constraint error violates Ineq. (16-1), the algorithm starts with a restoration phase. Whenever the algorithm starts with a restoration phase (in practice, this occurrence is quite common), the first cycle is a half-cycle or incomplete cycle, since it does not include a gradient phase.

Remark 4.4. At the end of the gradient phase of any cycle, the constraint error P should be computed. If P violates Ineq. (16-1), the gradient phase is followed by a restoration phase. If P satisfies Ineq. (16-1), the gradient phase is followed by another gradient phase, that is, the restoration phase is bypassed.

Remark 4.5. If the constraints (2)–(4) are linear in x , u , π , the initial restoration phase leads to complete constraint satisfaction, providing it is executed with a stepsize $\alpha = 1$. Subsequently, the restoration phase is bypassed, since the gradient phase does not produce any constraint violation. This is the same as stating that any subsequent gradient-restoration cycle is composed of a gradient phase only.

Remark 4.6. In both the gradient phase and the restoration phase, a linear, two-point boundary-value problem must be solved. Once the constants k_i are known, the composite solution is obtained via (43)–(44). A drawback of (43)–(44) is that $q + 1$ particular solutions $A_i(t)$, $B_i(t)$, $\lambda_i(t)$ must be stored at $N + 1$ time stations (here, N denotes the number of integration subintervals, so that $\Delta t = 1/N$ is the magnitude of the integration step). Hence, a storage problem arises if the system (2)–(4) is relatively large and if the computer memory is relatively limited.

This drawback can be offset as follows. Once the constants k_i are known, the multiplier μ of the composite solution is computed with (44-2), which yields

$$\mu = [k_1, k_2, \dots, k_q]^T. \quad (67)$$

Then, a supplementary sweep is executed according to (a) through (f) of Section 3.5. Clearly, the total number of sweeps increases from $q + 1$ to $q + 2$; consequently, the computational work per iteration, gradient or restorative, becomes proportional to the factor

$$w = 2n(q + 2). \quad (68)$$

In conclusion, use of this supplementary sweep increases the CPU time; nevertheless, depending on the severity of the storage problem, this course of action might still be desirable, and sometimes essential, with certain systems and certain computers.

Remark 4.7. For the gradient phase, the following relation can be shown to hold between the isoperimetric constant K and the stepsize α :

$$K = Q\alpha^2, \quad (69)$$

where

$$Q = \int_0^1 B^T B \, dt + C^T C. \quad (70)$$

Once the two-point boundary-value problem (34)–(40) has been solved, the functions $B(t)$ and C are known, and the error in the optimality conditions can be computed with (70). Therefore, (69) establishes a correspondence between the isoperimetric constant K and the value of the stepsize α . However, there is no way to determine *a priori* convenient values for the isoperimetric constant K . Therefore, the implementation of the algorithm becomes simpler if one avoids evaluating α in terms of K and assigns value to α directly. This is precisely the point of view taken in Section 3.9.

5. Combined Gradient-Restoration Algorithm

In the previous section, we discussed at length the sequential gradient-restoration algorithm. As stated, this algorithm includes the alternate succession of gradient phases and restoration phases. In the gradient phase, the first variation property (25-1) holds; hence, the total variation property (24-1) can be enforced. In the restoration phase, the first variation property (25-2) holds; hence, the total variation property (24-2) can be enforced.

In this section, we consider a first modification of the algorithm given in Section 3. This is the combined gradient-restoration algorithm, an iterative algorithm in which the gradient phase and the restoration phase are combined in a single phase, called the *gradient-restoration phase*. To do so, the first variation properties (25-1) and (25-2) must hold simultaneously; hence, the total variation properties (24-1) and (24-2) can be enforced simultaneously, if so desired (Ref. 65).

In order to generate variations which simultaneously lower the augmented functional and the constraint error, we consider the following auxiliary minimization problem.

Problem (A-4). For the gradient-restoration phase, minimize the functional

$$\delta I = \int_0^1 \delta f dt + (\delta g)_1, \quad (71)$$

with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) + \alpha(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (72)$$

$$\delta(x)_0 = 0, \quad (73)$$

$$\delta(\psi)_1 + \alpha(\psi)_1 = 0, \quad (74)$$

and the isoperimetric constraint

$$K = \int_0^1 \Delta u^T \Delta u dt + \Delta \pi^T \Delta \pi. \quad (75)$$

Clearly, Problem (A-4) is a variational problem of the Bolza type. Upon deriving the optimality conditions and invoking the definitions (22), we see that the set of linearized constraints and optimality conditions takes once more the form represented by Eqs. (34)–(40), with the following provision: the constants k_g and k_r appearing in (34)–(40) assume the values:

$$\text{gradient-restoration phase, } k_g = 1, \quad k_r = 1. \quad (76)$$

The solution of the linear, two-point boundary-value problem (34)–(40) is described in Section 3.5 and is not repeated here. Associated with the resulting functions $A(t)$, $B(t)$, C are the following first variation properties:⁶

$$\delta J = -\alpha Q, \quad \delta P = -2\alpha P, \quad (77)$$

consistent with (25-1) and (25-2).

Next, we consider the one-parameter family of solutions

$$\tilde{x}(t) = x(t) + \alpha A(t), \quad \tilde{u}(t) = u(t) + \alpha B(t), \quad \tilde{\pi} = \pi + \alpha C, \quad (78)$$

where α is the parameter. In principle, the stepsize α can be determined so that the total variation properties (24-1) and (24-2) are enforced simultaneously, that is,

$$\tilde{J}(\alpha) < \tilde{J}(0), \quad \tilde{P}(\alpha) \leq \tilde{P}(0). \quad (79)$$

⁶ Note that Q is given by Eq. (70).

In practice, the simultaneous enforcement of (79-1) and (79-2) is not desirable. This can be understood by considering the special situation where the nominal functions $x(t)$, $u(t)$, π satisfy the constraints (2)–(4) exactly. If the constraints are nonlinear, the only value of the stepsize compatible with (79-2) is $\alpha = 0$; hence, the combined gradient–restoration algorithm is unable to produce varied functions which are different in any way from the nominal functions.

In the light of the above considerations, it is prudent to enforce Ineq. (79-1) only and either avoid using Ineq. (79-2) or employ a modification of it. The modification which makes sense is the one designed to prevent excessive constraint violation as well as to guarantee the existence of a finite value for the optimal stepsize. With this in mind, Ineqs. (79) are replaced by

$$\tilde{J}(\alpha) < \tilde{J}(0), \quad \tilde{P}(\alpha) \leq \tilde{P}(0) + P_*. \quad (80)$$

It is interesting to note that Ineqs. (80) reduce to Ineqs. (52) if the nominal functions $x(t)$, $u(t)$, π satisfy the constraints (2)–(4) exactly.

6. Family of Gradient-Restoration Algorithms

In the previous sections, two basic algorithms for solving Problem (A-1) were considered: the sequential gradient–restoration algorithm (SGRA) and the combined gradient–restoration algorithm (CGRA). In this section, several algorithms which can be generated from the two basic ones through simple modifications are considered. We summarize the terminology employed to designate the elements of this family of gradient–restoration algorithms as follows (Ref. 67):

- (i) SGRA-CR, sequential gradient–restoration algorithm, with complete restoration;
- (ii) SGRA-AR, sequential gradient–restoration algorithm, with alternate restoration;
- (iii) SGRA-OR, sequential gradient–restoration algorithm, with optional restoration;
- (iv) CGRA-NR, combined gradient–restoration algorithm, with no restoration;
- (v) CGRA-AR, combined gradient–restoration algorithm, with alternate restoration;
- (vi) CGRA-OR, combined gradient–restoration algorithm, with optional restoration.

6.1. Description of the Algorithms. All of the previous algorithms are governed by Eqs. (34)–(40) and differ from one another through different specifications of the constants k_g and k_r . The detailed structure of these algorithms is given below.

SGRA-CR. This is precisely the algorithm described in Section 3. The decision parameter is the constraint error P , defined by (11). If $P \leq \epsilon_1$, the algorithm is in gradient phase; if $P > \epsilon_1$, the algorithm is in restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (41) for the gradient phase and (42) for the restoration phase. The stepsize search must be subordinated to Ineqs. (52) for the gradient phase and Ineq. (53) for the restoration phase.

SGRA-AR. This is a modification of the algorithm described in Section 3. The decision parameter is the constraint error P . If $P \leq \epsilon_1$, the algorithm is in gradient phase. If $P > \epsilon_1$, the algorithm executes alternately a gradient phase and a restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (41) for the gradient phase and (42) for the restoration phase. The stepsize search must be subordinated to Ineqs. (80) for the gradient phase and Ineq. (53) for the restoration phase.

SGRA-OR. This is a further modification of the algorithm described in Section 3. The decision parameter is the quantity

$$Z = P/Q^\beta, \quad (81)$$

where P is the constraint error (11) and Q is the error in the optimality conditions (12). For example, the exponent β can be defined by

$$\beta = 1 \quad \text{if } P > 1 \quad \text{and/or} \quad Q > 1, \quad (82)$$

$$\beta = (\log_{10} \epsilon_1) / (\log_{10} \epsilon_2) \quad \text{if } P \leq 1 \quad \text{and} \quad Q \leq 1, \quad (83)$$

where ϵ_1 and ϵ_2 denote the preselected tolerances for P and Q . As an example, for

$$\epsilon_1 = 10^{-8}, \quad \epsilon_2 = 10^{-4}, \quad (84)$$

the decision parameter (81) becomes

$$Z = P/Q \quad \text{if } P > 1 \quad \text{and/or} \quad Q > 1, \quad (85)$$

$$Z = P/Q^2 \quad \text{if } P \leq 1 \quad \text{and} \quad Q \leq 1. \quad (86)$$

If $Z \leq 1$, the algorithm is in gradient phase. If $Z > 1$, the algorithm is in restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (41) for the gradient phase and (42) for the restoration phase. The stepsize search must be subordinated to Ineqs. (80) for the gradient phase and Ineq. (53) for the restoration phase.

CGRA-NR. This is precisely the algorithm described in Section 5. There is no decision parameter, and all of the phases have an identical structure, that of a combined gradient–restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (76). The stepsize search must be subordinated to Ineqs. (80).

CGRA-AR. This is a modification of the algorithm described in Section 5. The decision parameter is the constraint error P . If $P \leq \epsilon_1$, the algorithm is in gradient–restoration phase. If $P > \epsilon_1$, the algorithm executes alternately a gradient–restoration phase and a restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (76) for the gradient–restoration phase and (42) for the restoration phase. The stepsize search must be subordinated to Ineqs. (80) for the gradient–restoration phase and Ineq. (53) for the restoration phase.

CGRA-OR. This is a further modification of the algorithm described in Section 5. The decision parameter is the quantity Z , defined by (81)–(83). If $Z \leq 1$, the algorithm is in gradient–restoration phase. If $Z > 1$, the algorithm is in restoration phase.

When solving the system (34)–(40), the characteristic constants k_g and k_r take the values (76) for the gradient–restoration phase and (42) for the restoration phase. The stepsize search must be subordinated to Ineqs. (80) for the gradient–restoration phase and Ineq. (53) for the restoration phase.

6.2. Summary of the Algorithms. The main characteristics of the previous algorithms are illustrated graphically in Figs. 1–6. Specifically, Fig. 1 refers to SGRA-CR, Fig. 2 refers to SGRA-AR, and Fig. 3 refers to SGRA-OR. Also, Fig. 4 refers to CGRA-NR, Fig. 5 refers to CGRA-AR, and Fig. 6 refers to CGRA-OR.

Obviously, Figs. 1–6 must be interpreted in an abstract way. An arbitrary point in Figs. 1–6 represents some infeasible combination of

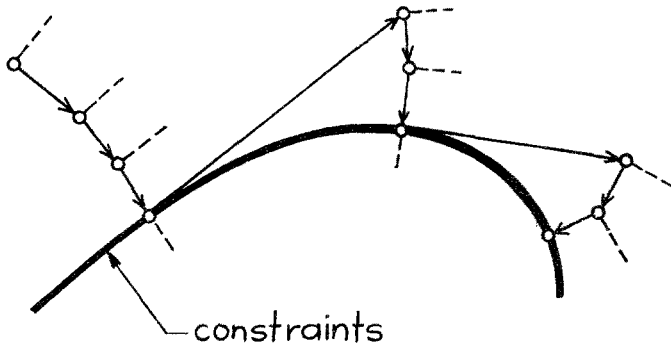


Fig. 1. Sequential gradient-restoration algorithm, with complete restoration (SGRA-CR).

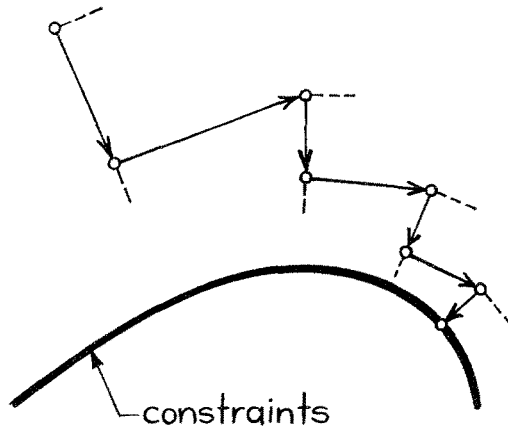


Fig. 2. Sequential gradient-restoration algorithm, with alternate restoration (SGRA-AR).

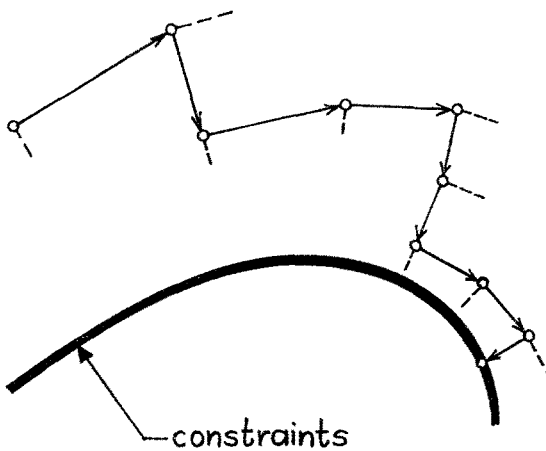


Fig. 3. Sequential gradient-restoration algorithm, with optional restoration (SGRA-OR).

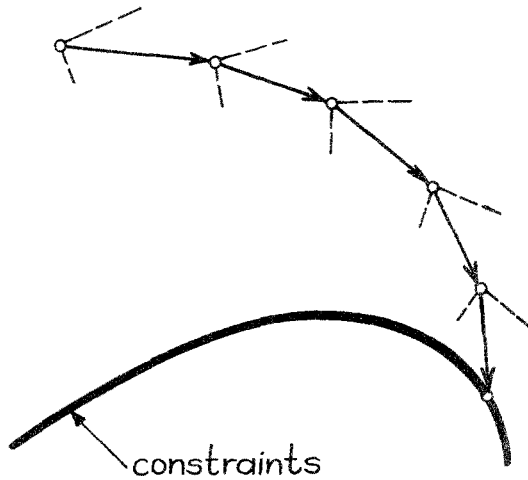


Fig. 4. Combined gradient-restoration algorithm, with no restoration (CGRA-NR).

state $x(t)$, control $u(t)$, and parameter π . However, a point located on the line labeled with the word *constraints* represents a feasible combination of state, control, and parameter.

Among the elements of the family of algorithms considered, only SGRA-CR produces a succession of feasible suboptimal solutions (Fig. 1); hence, for this algorithm, feasibility is reached long before optimality. The remaining algorithms do not produce a succession of

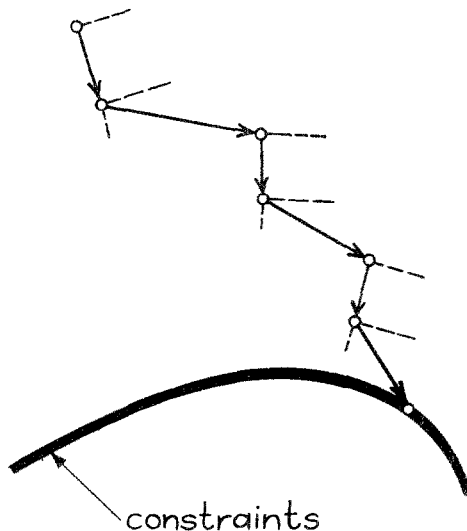


Fig. 5. Combined gradient-restoration algorithm, with alternate restoration (CGRA-AR).

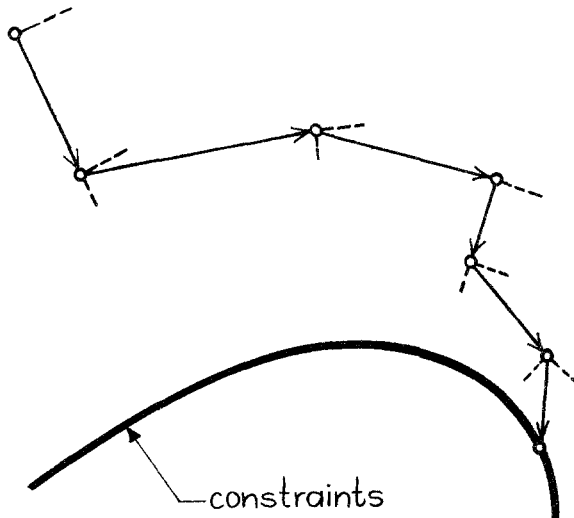


Fig. 6. Combined gradient-restoration algorithm, with optional restoration (CGRA-OR).

feasible suboptimal solutions (Figs. 2-6); hence, for these algorithms, feasibility is reached sometimes before, sometimes simultaneously with, and sometimes after optimality, depending on the particular problem and the choice of the nominal functions.

6.3. Computational Considerations. Assume now that the method of particular solutions is employed in solving the linear, two-point boundary-value problem (34)-(40). Recall that each iteration requires integrating $2n$ differential equations $q + 1$ times; hence, the computational effort per iteration is proportional to the factor

$$w = 2n(q + 1). \quad (87)$$

Since this factor is common to all of the previous algorithms, the computer time per iteration is approximately the same. Hence, a comparison of the previous algorithms based on the number of iterations for convergence makes sense (Ref. 67).

6.4. Computational Experience. Extensive computer experimentation has been carried out by this author and his associates not only on the above family of algorithms (Refs. 63 and 67), but also on their counterparts in mathematical programming. A wide variety of sample problems has been solved, and the number of iterations for convergence has been recorded. The extensive analyses have led to remarkably consistent results and to these conclusions:

(i) the inclusion of a separate restoration phase is essential to good performance of gradient-restoration algorithms;

(ii) SGRA-CR is the most stable among the totality of algorithms considered here; conversely, CGRA-NR is the least stable;

(iii) occasionally, rapidity of convergence can be increased somewhat if one employs algorithms with alternate restoration or optional restoration.

Based on the previous results, the author's personal preference is for SGRA-CR, the sequential gradient-restoration algorithm, with complete restoration. Not only is it the most stable among the algorithms considered here, but it is the only one, among the several algorithms considered, which produces a succession of feasible suboptimal solutions. This is because Ineq. (16-1) is enforced at the beginning as well as the end of each complete gradient-restoration cycle.

It is clear that the availability of a succession of feasible suboptimal solutions is important to an engineer or a physicist. It can happen that a solution optimizing a given functional I_1 may be undesirable from the point of view of the value of some other functional I_2 which is equally important within the context of the problems being considered. If the algorithm employed, because of its nature, produces feasible suboptimal solutions, then the way is open for a trade-off between the optimal solution and some of the suboptimal solutions calculated along the way to convergence. For example, the trade-off can be based on the considerations of the following linear combination of functionals:

$$I = k_1 I_1 + k_2 I_2, \quad (88)$$

where k_1 and k_2 are preassigned positive constants such that

$$k_1 + k_2 = 1. \quad (89)$$

7. Hybrid Modification of the Sequential Gradient-Restoration Algorithm

In this section, we present a modification of the sequential gradient-restoration algorithm, designed to improve the control distribution and, hence, the convergence characteristics (Ref. 66). We call this a *hybrid modification*, because it has characteristics intermediate between those of a first-order method and those of a second-order method.

As stated before, the sequential gradient-restoration algorithm of Section 3 includes a gradient phase, governed by Problem (A-2), and a restoration phase, governed by Problem (A-3). In the present hybrid modification, the restoration phase is still governed by Problem (A-3). On the other hand, the gradient phase is obtained by replacing Problem (A-2) with the following auxiliary minimization problem.

Problem (A-5). For the gradient phase, minimize the functional (26) with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints (27)–(29) and the quadratic isoperimetric constraint

$$K = \int_0^1 \Delta u^T W \Delta u \, dt + \Delta \pi^T U \Delta \pi, \quad (90)$$

where $W(t)$ and U are symmetric, positive-definite weighting matrices having dimensions $m \times m$ and $p \times p$, respectively.

7.1. Determination of the Basic Functions. With $W(t)$ and U specified *a priori*, the basic functions $A(t)$, $B(t)$, C must be obtained by solving the following linear, two-point boundary-value problem:

$$\dot{A} - \varphi_x^T A - \varphi_u^T B - \varphi_\pi^T C + k_r(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (91)$$

$$(A)_0 = 0, \quad (92)$$

$$(\psi_x^T A + \psi_\pi^T C + k_r \psi)_1 = 0, \quad (93)$$

and

$$\dot{\lambda} - k_g f_x + \varphi_x \lambda = 0, \quad 0 \leq t \leq 1, \quad (94)$$

$$(\lambda + k_g g_x + \psi_x \mu)_1 = 0, \quad (95)$$

$$B + W^{-1}(k_g f_u - \varphi_u \lambda) = 0, \quad 0 \leq t \leq 1, \quad (96)$$

$$C + U^{-1} \left[\int_0^1 (k_g f_\pi - \varphi_\pi \lambda) \, dt + (k_g g_\pi + \psi_\pi \mu)_1 \right] = 0. \quad (97)$$

In (91)–(97), the constants k_g and k_r take the values (41) for the gradient phase and (42) for the restoration phase. For the restoration phase, the weighting matrices $W(t)$ and U are chosen to be

$$W = I_m, \quad U = I_p, \quad (98)$$

where I_m and I_p denote the identity matrices of order m and p , respectively. Therefore, for the restoration phase, the system (91)–(97) reduces

to (34)–(40). For the gradient phase, the specification of the weighting matrices is discussed below.

7.2. Specification of the Weighting Matrices. For the gradient phase, it is tempting to specify the weighting matrices appearing in (91)–(97) as follows:

$$W = H_{uu}, \quad U = G_{\pi\pi}, \quad (99)$$

where H and G denote the expressions⁷

$$H = f - \lambda^T \varphi, \quad 0 \leq t \leq 1, \quad (100)$$

$$G = \int_0^1 (f - \lambda^T \varphi) dt + (g + \mu^T \psi)_1. \quad (101)$$

With these specifications, one hopes to arrive at a version of the sequential-gradient restoration algorithm which partakes of some of the characteristics of a quasilinearization algorithm.

It is readily apparent that the choices (99) must be subject to changes dictated by practical considerations. This can be seen by combining (99)–(101) to yield

$$W = f_{uu} - \varphi_{uu} \lambda, \quad 0 \leq t \leq 1, \quad (102)$$

$$U = \int_0^1 (f_{\pi\pi} - \varphi_{\pi\pi} \lambda) dt + (g_{\pi\pi} + \psi_{\pi\pi} \mu)_1. \quad (103)$$

While the matrices f_{uu} , $f_{\pi\pi}$, $g_{\pi\pi}$ and the three-dimensional arrays φ_{uu} , $\varphi_{\pi\pi}$, $\psi_{\pi\pi}$ are known *a priori*, the multipliers $\lambda(t)$ and μ are not known *a priori*. Therefore, the choice (102)–(103) for the weighting matrices would cause the boundary-value problem (91)–(97) to become nonlinear. Now, it is essential to avoid nonlinearities within the iteration; hence, we replace (102) and (103) with

$$W = f_{uu} - \varphi_{uu} \hat{\lambda}, \quad 0 \leq t \leq 1, \quad (104)$$

$$U = \int_0^1 (f_{\pi\pi} - \varphi_{\pi\pi} \hat{\lambda}) dt + (g_{\pi\pi} + \psi_{\pi\pi} \hat{\mu})_1, \quad (105)$$

where $\hat{\lambda}(t)$ and $\hat{\mu}$ denote the multipliers associated with the previous gradient iteration. With this change, the matrices $W(t)$ and U are now known *a priori*, and the linearity of the boundary-value problem (91)–(97) is preserved.

⁷ The symbol H denotes the so-called Hamiltonian of the system governed by (1)–(4).

When employing the weighting matrices (104)–(105) in conjunction with (91)–(97), several supplementary precautions must be taken.

(i) The matrix $W(t)$ must be positive definite everywhere along the interval of integration. Otherwise, $W(t)$ must be replaced by the identity matrix I_m either everywhere or for those time subintervals where $W(t)$ is not positive definite.

(ii) The matrix U must be positive definite. Otherwise, it must be replaced by the identity matrix I_p .

(iii) With reference to (ii), an important particular case is the minimum-time problem for an autonomous system. If the final time τ is the only parameter and one sets $\pi = \tau$, it is clear that G is linear in π , so that $G_{\pi\pi} = 0$. Therefore, the matrix U must be set equal to the identity matrix I_p .

(iv) For the first gradient phase, the multipliers $\hat{\lambda}(t)$ and $\hat{\mu}$ are nonexistent. Hence, the matrices $W(t)$ and U must be set equal to the identity matrices I_m and I_p , respectively.

7.3. Descent Properties. With the above precautions in mind, the functions $A(t)$, $B(t)$, C solving Eqs. (91)–(97) can be shown to have the following first variation properties (Ref. 66):

$$\text{gradient phase, } \delta J = -\alpha \left[\int_0^1 B^T W B \, dt + C^T U C \right]; \quad (106)$$

$$\text{restoration phase, } \delta P = -2\alpha P. \quad (107)$$

As a consequence, the stepsize α of the one-parameter family of solutions (50) can be determined by enforcing Ineqs. (52) for the gradient phase and Ineq. (53) for the restoration phase.

7.4. Computational Experience. Some computational experience on this hybrid modification of the sequential gradient–restoration algorithm has been reported in Ref. 66. It shows that this hybrid modification exhibits better convergence characteristics [namely, a smaller number of iterations required to achieve the stopping conditions (16)] than the sequential gradient–restoration algorithm of Section 3. It must be realized, however, that this smaller number of iterations for convergence is obtained at the expense of a larger computer time per iteration, in view of the need of (a) memorizing the multipliers $\hat{\lambda}(t)$ and $\hat{\mu}$ of the previous gradient iteration, (b) computing certain matrices and three-dimensional arrays everywhere along the interval of integration, and (c) checking the positive-definiteness of the weighting matrices $W(t)$ and U .

8. Sequential Conjugate Gradient-Restoration Algorithm

In the previous section, we described a hybrid modification of SGRA-CR, designed to accelerate terminal convergence. This hybrid modification has characteristics partaking of both a first-order method and a second-order method.

In this section, we describe another modification of SGRA-CR, designed to accelerate terminal convergence, while retaining the characteristics of a first-order method. This is the conjugate gradient version of the sequential gradient-restoration algorithm (Refs. 72-73).

As stated before, SGRA-CR includes a gradient phase, governed by Problem (A-2), and a restoration phase, governed by Problem (A-3). In the conjugate gradient modification, the restoration phase is still governed by Problem (A-3). On the other hand, the conjugate gradient phase is obtained by replacing Problem (A-2) with the following auxiliary minimization problem.

Problem (A-6). For the conjugate gradient phase, minimize the functional (26) with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints (27)-(29) and the quadratic isoperimetric constraint

$$K = \int_0^1 (\Delta u - \beta \Delta \hat{u})^T (\Delta u - \beta \Delta \hat{u}) dt + (\Delta \pi - \beta \Delta \hat{\pi})^T (\Delta \pi - \beta \Delta \hat{\pi}). \quad (108)$$

In Problem (A-6), β denotes a scalar parameter, yet to be specified; $\Delta x(t)$, $\Delta u(t)$, $\Delta \pi$ denote the variations of state, control, and parameter pertaining to the present conjugate gradient phase; and $\Delta \hat{x}(t)$, $\Delta \hat{u}(t)$, $\Delta \hat{\pi}$ denote the variations of state, control, and parameter pertaining to the previous conjugate gradient phase. Clearly, the solution of the auxiliary minimization problem is sought within the class of variations of the control and the parameter which are equidistant from some constant multiple of their predecessors.

8.1. Determination of the Basic Functions. With $\hat{A}(t)$, $\hat{B}(t)$, $\hat{C}$ specified *a priori*, the basic functions $A(t)$, $B(t)$, C must be obtained by solving the following linear, two-point boundary-value problem:

$$\dot{A} - \varphi_x^T A - \varphi_u^T B - \varphi_\pi^T C + k_r(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (109)$$

$$(A)_0 = 0, \quad (110)$$

$$(\psi_x^T A + \psi_\pi^T C + k_r \psi)_1 = 0, \quad (111)$$

and

$$\lambda - k_g f_x + \varphi_x \lambda = 0, \quad 0 \leq t \leq 1, \quad (112)$$

$$(\lambda + k_g g_x + \psi_x \mu)_1 = 0, \quad (113)$$

$$B - k_g \gamma \hat{B} + k_g f_u - \varphi_u \lambda = 0, \quad 0 \leq t \leq 1, \quad (114)$$

$$C - k_g \gamma \hat{C} + \int_0^1 (k_g f_\pi - \varphi_\pi \lambda) dt + (k_g g_\pi + \psi_\pi \mu)_1 = 0, \quad (115)$$

where

$$\gamma = \beta(\hat{\alpha}/\alpha) \quad (116)$$

is the directional coefficient. In (109)–(115), the constants k_g and k_r take the values (41) for the gradient phase and (42) for the restoration phase. For the restoration phase, the system (109)–(115) reduces to (34)–(40). For the gradient phase, the specification of the directional coefficient γ is discussed below.

8.2. Linear-Quadratic Case. We consider now the special case where the functions f , g appearing in the functional (1) are quadratic in x , u , π and the functions φ , ψ appearing in the constraints (2)–(4) are linear in x , u , π . For this special case, the solution of the system (109)–(115) can be written in the form (Ref. 72)

$$A(t) = A_0(t) + \gamma \hat{A}(t), \quad B(t) = B_0(t) + \gamma \hat{B}(t), \quad C = C_0 + \gamma \hat{C}, \quad (117)$$

$$\lambda(t) = \lambda_0(t), \quad \mu = \mu_0 \quad (118)$$

In (117)–(118), $A(t)$, $B(t)$, C , $\lambda(t)$, μ denote the complete solutions of (109)–(115) corresponding to the present conjugate gradient iteration; and $\hat{A}(t)$, $\hat{B}(t)$, $\hat{C}$, $\hat{\lambda}(t)$, $\hat{\mu}$ denote the complete solutions of (109)–(115) corresponding to the previous conjugate gradient iteration. On the other hand, $A_0(t)$, $B_0(t)$, C_0 , $\lambda_0(t)$, μ_0 denote the particular solutions of (109)–(115) corresponding to $\gamma = 0$; obviously, these particular solutions are identical with the solutions of (34)–(40), that is, they are the solutions of the linear, two-point boundary-value problem associated with the ordinary gradient method of Section 3.

If we combine (117) with (23), we obtain a two-parameter family of solutions having the form

$$\tilde{x} = \tilde{x}(\alpha, \gamma), \quad \tilde{u} = \tilde{u}(\alpha, \gamma), \quad \tilde{\pi} = \tilde{\pi}(\alpha, \gamma), \quad (119)$$

for which the augmented functional (5) becomes a function of the form

$$\tilde{J} = \tilde{J}(\alpha, \gamma). \quad (120)$$

The optimal values of the parameters are then obtained by minimizing (120) with respect to α and γ , that is, by solving the relations

$$\bar{J}_\alpha(\alpha, \gamma) = 0, \quad \bar{J}_\gamma(\alpha, \gamma) = 0. \quad (121)$$

It can be shown that the solution for α depends on the Hessian of the function f with respect to x, u, π as well as on the Hessian of the function g with respect to x, π . Since second derivatives must be avoided in a first-order method, this solution is abandoned in favor of a precise one-dimensional search (see below).

On the other hand, the solution for γ does not depend on second derivatives and can be expressed in terms of the solutions of (109)–(115) corresponding to $\gamma = 0$. Laborious manipulations, omitted for the sake of brevity, lead to the following optimal value for the directional coefficient:

$$\gamma = Q/\hat{Q}, \quad (122)$$

where

$$Q = \int_0^1 B_0^T B_0 \, dt + C_0^T C_0, \quad (123)$$

$$\hat{Q} = \int_0^1 \hat{B}_0^T \hat{B}_0 \, dt + \hat{C}_0^T \hat{C}_0. \quad (124)$$

In (122)–(124), Q is the error in the optimality conditions at the beginning of the present conjugate gradient phase and $\hat{Q}$ is the error in the optimality conditions at the beginning of the previous conjugate gradient phase.

For the linear-quadratic case, the functions appearing in (117) satisfy some general orthogonality conditions and conjugacy conditions, omitted here for the sake of brevity. In addition, the augmented functional (5) has the first variation property

$$\delta J = -\alpha Q, \quad (125)$$

so that the descent property (52-1) can be enforced.

The stepsize α must be determined through a precise search, which is essential to the preservation of the general orthogonality and conjugacy conditions mentioned above. For example, single-step quadratic interpolation can be employed (see Section 3.9) with guaranteed satisfaction of the stopping conditions (58).

In closing, the following remarks are pertinent.

(i) For the linear-quadratic case, only one restoration phase is needed at the very beginning of the algorithm. Once the constraints are reached (after a single restorative iteration, with $\alpha = 1$), they are never violated afterward. Hence, every subsequent iteration is a conjugate gradient iteration.

(ii) For the first conjugate gradient phase, the functions $\hat{A}(t)$, $\hat{B}(t)$, $\hat{C}$ are nonexistent. Therefore, one sets

$$\hat{A}(t) = 0, \quad \hat{B}(t) = 0, \quad \hat{C} = 0, \quad (126)$$

with the implication that the first conjugate gradient phase is actually an ordinary gradient phase. Alternatively, one can assign arbitrary values to $\hat{A}(t)$, $\hat{B}(t)$, $\hat{C}$ and set

$$\gamma = 0. \quad (127)$$

8.3. General Case. We consider now the case where the functions f , g are nonquadratic in x , u , π and/or the functions φ , ψ are nonlinear in x , u , π . For this general case, Eqs. (117)–(118) can be retained, with certain precautions taken. The directional coefficient γ is still given by (122)–(124). The stepsize α must be computed by a precise one-dimensional search. For example, multi-step cubic interpolation can be employed (see Section 3.9) until satisfaction of one of the stopping conditions (58) occurs.

With $A(t)$, $B(t)$, C determined as above, the first variation property (125) must be replaced by

$$\delta J = -\alpha Q - \alpha\gamma \left[\int_0^1 B_0^T \hat{B} \, dt + C_0^T \hat{C} \right]. \quad (128)$$

Therefore, it is not automatically clear that the descent property (52-1) can be enforced.

For the first conjugate gradient phase ($\gamma = 0$), Eq. (128) reduces to (125), so that the descent property (52-1) can be enforced. For subsequent iterations ($\gamma \neq 0$), the negativeness of (128) may or may not hold. Whenever $\delta J > 0$, then the conjugate gradient phase must be interrupted; this means that the conjugate gradient cycle must be restarted with an ordinary gradient phase, characterized by $\gamma = 0$.

8.4. Computational Experience. Some computational experience on SCGRA (sequential conjugate gradient–restoration algorithm) has been reported in Ref. 73. It shows that SCGRA generally exhibits better convergence characteristics [namely, a smaller number of iterations required to achieve the stopping conditions (16)] than SOGRA (sequential ordinary gradient–restoration algorithm).⁸ It must be realized, however, that this smaller number of iterations for convergence is obtained at the expense of a larger computer time per iteration, in view of the need of (a) memorizing the functions $\hat{A}(t)$, $\hat{B}(t)$, $\hat{C}$ and the multipliers $\hat{\lambda}(t)$, $\hat{\mu}$ of

⁸ Within the context of this section, the symbol SOGRA denotes the algorithm of Section 3.

the previous conjugate gradient iteration and (b) performing an accurate multi-step search in the determination of the conjugate gradient stepsize.

At any rate, the experiments of Ref. 73 show that the advantage of SCGRA with respect to SOGRA depends strongly on the problem being solved. It is higher for problems with free endpoints than for problems with fixed endpoints. It is also higher for linear-quadratic problems than for nonlinear-nonquadratic problems.

For linear-quadratic problems of optimal control, there seems to be a clear advantage in the use of conjugate gradient techniques. On the other hand, for nonlinear-nonquadratic problems, the advantage of conjugate gradient techniques with respect to ordinary gradient techniques is not as clear cut as for linear-quadratic problems.

The *probable* explanation for this behaviour is as follows. Essentially, an optimal control problem can be viewed to be equivalent to a mathematical programming problem with a large number of degrees of freedom. In turn, the number of degrees of freedom is of the same order as the product of the number of integration steps times the number of independent controls. For a problem where the number of integration steps is 100, the number of degrees of freedom is of order 100. This means that, for the conjugate gradient properties to become manifest, the number of iterations would have to be of order 100, which is a prohibitive number for problems of optimal control. Usually, both ordinary and conjugate gradient techniques enable one to achieve convergence in less than 100 iterations if the tolerance level set for the error in the optimality conditions is not excessively small.

It must be noted that the previous conclusions concern the number of iterations for convergence and not the computation time (CPU time). If one measures computer time, then the conclusions remain unchanged for the linear-quadratic case. For the nonlinear-nonquadratic case, the conclusions become somewhat less favorable to conjugate gradient techniques, since each iteration of SCGRA requires a larger computer time than each iteration of SOGRA.

9. Nondifferential Constraints

We continue our review by considering the following modification of Problem (A-1).

Problem (B-1). Minimize the functional

$$I = \int_0^1 f(x, u, \pi, t) dt + [g(x, \pi, t)]_1, \quad (129)$$

with respect to the state $x(t)$, the control $u(t)$, and the parameter π which satisfy the constraints

$$\dot{x} - \varphi(x, u, \pi, t) = 0, \quad 0 \leq t \leq 1, \quad (130)$$

$$S(x, u, \pi, t) = 0, \quad 0 \leq t \leq 1, \quad (131)$$

$$(x)_0 = \text{given}, \quad (132)$$

$$[\psi(x, \pi, t)]_1 = 0. \quad (133)$$

In Eqs. (129)–(133), the quantities I, f, g, t are scalar, and the quantities $x, u, \pi, \varphi, \psi, S$ are vectors having dimensions n, m, p, n, q, r , respectively.

Problem (B-1) differs from Problem (A-1) of Section 2 because of the presence of the nondifferential constraint (131) to be satisfied everywhere along the interval of integration. In theory, this seems to be only a minor change. In practice, the change is considerable in that it enlarges dramatically the number and variety of problems of optimal control which can be treated by gradient–restoration algorithms. Indeed, by suitable transformations, almost every known problem of optimal control theory can be brought into the scheme considered here (Ref. 71). This statement applies, for instance, to the following situations: (i) problems with control equality constraints, (ii) problems with state equality constraints, (iii) problems with equality constraints on the time rate of change of the state, (iv) problems with control inequality constraints, (v) problems with state inequality constraints, and (vi) problems with inequality constraints on the time rate of change of the state. For an illustration of the scope and range of applicability of the present formulation, the reader is referred to Refs. 31–33, where a total of 24 numerical examples is given.

Remark. Occasionally, the state x and/or the parameter π can be absent from Eq. (131). However, this can never be the case with the control u , whose presence in *each* of the scalar components of Eq. (131) is postulated. Therefore, it is assumed that suitable transformations are introduced, if necessary, in order to convert problems where the function S does not involve the control into problems where the function S involves the control (see Ref. 71 and Refs. 31–33).

9.1. First-Order Conditions. The previous Problem (B-1) is one of the Bolza type and is characterized by the augmented functional

$$J = \int_0^1 [f + \lambda^T(\dot{x} - \varphi) + \rho^T S] dt + (g + \mu^T \psi)_1, \quad (134)$$

where λ, ρ, μ are vector Lagrange multipliers having dimensions n, r, q , respectively. It is known that the minimization of (129), subject to

(130)–(133), is identical with the minimization of (134), subject to (130)–(133), regardless of the choice of the multipliers $\lambda(t)$, $\rho(t)$, μ .

If the multipliers $\lambda(t)$, $\rho(t)$, μ are chosen consistently with

$$\lambda - f_x + \varphi_x \lambda - S_x \rho = 0, \quad 0 \leq t \leq 1, \quad (135)$$

$$(\lambda + g_x + \psi_x \mu)_1 = 0, \quad (136)$$

then the optimum solution $x(t)$, $u(t)$, π must satisfy the following relations:

$$f_u - \varphi_u \lambda + S_u \rho = 0, \quad 0 \leq t \leq 1, \quad (137)$$

$$\int_0^1 (f_\pi - \varphi_\pi \lambda + S_\pi \rho) dt + (g_\pi + \psi_\pi \mu)_1 = 0. \quad (138)$$

The system composed of the feasibility equations (130)–(133) and the optimality conditions (135)–(138) constitutes a nonlinear, two-point boundary-value problem in which the unknowns are the functions $x(t)$, $u(t)$, π and the multipliers $\lambda(t)$, $\rho(t)$, μ . Rather than solving this system directly, we shall solve it indirectly by means of a sequential gradient-restoration algorithm, with complete restoration (SGRA-CR), similar in construction to that of Section 3. Prior to doing so, we shall define convergence in a numerical sense through Ineqs. (16), where P and Q are now the scalar performance indexes associated with Problem (B-1). These performance indexes are given by

$$P = \int_0^1 N(\dot{x} - \varphi) dt + \int_0^1 N(S) dt + N(\psi)_1, \quad (139)$$

$$Q = \int_0^1 N(\lambda - f_x + \varphi_x \lambda - S_x \rho) dt + \int_0^1 N(f_u - \varphi_u \lambda + S_u \rho) dt \\ + N \left[\int_0^1 (f_\pi - \varphi_\pi \lambda + S_\pi \rho) dt + (g_\pi + \psi_\pi \mu)_1 \right] + N(\lambda + g_x + \psi_x \mu)_1 \quad (140)$$

and measure the errors in the constraints and the optimality conditions, respectively.

9.2. Sequential Gradient-Restoration Algorithm. In constructing a sequential gradient-restoration algorithm for Problem (B-1), we shall retain the properties of Section 3.1 and the notation of Section 3.2. We shall represent the varied functions $\tilde{x}(t)$, $\tilde{u}(t)$, $\tilde{\pi}$ through Eqs. (23). Therefore, we are faced with two problems: (a) the determination of the basic functions $A(t)$, $B(t)$, C and (b) the determination of the stepsize α .

9.3. Determination of the Basic Functions. The determination of the basic functions $A(t)$, $B(t)$, C is accomplished through the formulation of the following auxiliary minimization problems.

Problem (B-2). For the gradient phase, minimize the functional

$$\delta I = \int_0^1 \delta f \, dt + (\delta g)_1, \quad (141)$$

with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (142)$$

$$\delta(S) = 0, \quad 0 \leq t \leq 1, \quad (143)$$

$$\delta(x)_0 = 0, \quad (144)$$

$$\delta(\psi)_1 = 0, \quad (145)$$

and the isoperimetric constraint

$$K = \int_0^1 \Delta u^T \Delta u \, dt + \Delta \pi^T \Delta \pi. \quad (146)$$

Problem (B-3). For the restoration phase, minimize the functional (146), with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) + \alpha(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (147)$$

$$\delta(S) + \alpha S = 0, \quad 0 \leq t \leq 1, \quad (148)$$

$$\delta(x)_0 = 0, \quad (149)$$

$$\delta(\psi)_1 + \alpha(\psi)_1 = 0. \quad (150)$$

9.4. Linear, Two-Point Boundary-Value Problem. Problems (B-2) and (B-3) are variational problems of the Bolza type. The form of the linearized constraints and optimality conditions is such that the equations governing Problems (B-2) and (B-3) can be embedded in the following single system:

$$\dot{A} - \varphi_x^T A - \varphi_u^T B - \varphi_\pi^T C + k_r(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 1, \quad (151)$$

$$S_x^T A + S_u^T B + S_\pi^T C + k_r S = 0, \quad 0 \leq t \leq 1, \quad (152)$$

$$(A)_0 = 0, \quad (153)$$

$$(\psi_x^T A + \psi_\pi^T C + k_r \psi)_1 = 0, \quad (154)$$

and

$$\lambda - k_g f_x + \varphi_x \lambda - S_x \rho = 0, \quad 0 \leq t \leq 1, \quad (155)$$

$$(\lambda + k_g g_x + \psi_x \mu)_1 = 0, \quad (156)$$

$$B + k_g f_u - \varphi_u \lambda + S_u \rho = 0, \quad 0 \leq t \leq 1, \quad (157)$$

$$C + \int_0^1 (k_g f_\pi - \varphi_\pi \lambda + S_\pi \rho) \, dt + (k_g g_\pi + \psi_\pi \mu)_1 = 0. \quad (158)$$

The constants k_g and k_r appearing in (151)–(158) take the values (41)–(42).

For given nominal functions $x(t)$, $u(t)$, π and constants k_y and k_r , Eqs. (151)–(158) define the functions $A(t)$, $B(t)$, C and the multipliers $\lambda(t)$, $\rho(t)$, μ . Clearly, Eqs. (151)–(158) constitute a linear, two-point boundary-value problem, which can be solved independently of the value assigned to the stepsize α .

9.5. Method of Particular Solutions. The usual approach of backward integration of the λ -equation and forward integration of the A -equation (see Section 3.5) cannot be employed with this class of problems. This is due to the fact that the computation of the multiplier $\rho(t)$ over the interval $0 \leq t \leq 1$ requires the simultaneous integration of both the λ -equation and the A -equation. Because of the coupling, the total system must be integrated simultaneously in either the forward direction or the backward direction. Since either direction requires the same number of integrations, we choose the forward direction.

Let a *sweep* be defined as a forward integration of Eqs. (151)–(158), obtained by specifying the $n + p$ components of the vector

$$z = [\lambda^T(0), C^T]^T \quad (159)$$

and satisfying all of the equations, except (154), (156), and (158). Note that the successful execution of the integration is subordinated to the condition that

$$\det[S_u^T S_u] \neq 0, \quad 0 \leq t \leq 1. \quad (160)$$

This condition arises by observing that, at each time station t , and for given values of A , λ , and C , relations (152) and (157) constitute a system of $m + r$ equations in the $m + r$ components of the vectors B and ρ . This system admits a unique solution providing (160) is satisfied.

If the method of particular solutions is employed, one must execute $n + p + 1$ independent sweeps. Specifically, the first $n + p$ sweeps are performed by considering the identity matrix of order $n + p$ and by choosing the vector z associated with the i th integration, $i = 1, \dots, n + p$, as identical with the i th column of the above identity matrix. The last sweep is performed by choosing the vector z associated with the $(n + p + 1)$ th integration as identical with the null vector.

Next, we combine the particular solutions linearly and write the composite solution in the form

$$A(t) = \sum k_i A_i(t), \quad B(t) = \sum k_i B_i(t), \quad C = \sum k_i C_i, \quad (161)$$

$$\lambda(t) = \sum k_i \lambda_i(t), \quad \rho(t) = \sum k_i \rho_i(t), \quad (162)$$

where $i = 1, \dots, n + p + 1$. The $n + p + 1$ constants k_i are determined simultaneously with the q components of the multiplier μ by satisfying the linear system of order $n + p + q + 1$ composed of the n equations (156), the p equations (158), the q equations (154), and the supplementary scalar relation $\sum k_i = 1$, which arises from the nonhomogeneity of the system (151)–(158).

9.6. Computational Considerations. Each sweep involves integrating $2n$ differential equations, that is, the n linearized state equations (151) and the n multiplier equations (155). Since $n + p + 1$ sweeps are involved, the amount of computational work performed per iteration, gradient or restorative, is proportional to the factor

$$w = 2n(n + p + 1), \quad (163)$$

regardless of the number of final conditions q . Clearly, the computational effort per iteration increases with the square of the dimension of the state vector. Therefore, in devising transformations leading to problems of the form (129)–(133), one must keep in mind the following: whenever possible, it is a good idea to prevent or limit an increase in the dimension of the state vector x .

9.7. Determination of the Stepsize. The functions $A(t)$, $B(t)$, C solving Eqs. (151)–(158) are such that the first variation properties (48)–(49) hold. This being the case, it is possible to determine the stepsize α so that the descent property (24-1) is enforced for the gradient phase and the descent property (24-2) is enforced for the restoration phase.

Specifically, one has to consider the one-parameter family of solutions (50) for which the augmented functional (134) and the constraint error (139) take the form (51). Then, the stepsize α must be determined so that Ineqs. (52) are enforced for each gradient iteration and Ineq. (53) is enforced for each restorative iteration. To this effect, one can employ the search techniques outlined in Sections 3.9 and 3.10.

An order-of-magnitude analysis shows that (60)–(61) hold for the present problem. Hence, by a suitable choice of the gradient stepsize, the fundamental stability property (59) can be enforced for each complete gradient–restoration cycle.

9.8. Generalizations and Extensions. For Problem (B-1), generalizations and extensions similar to those shown for Problem (A-1) are possible. Specifically, one can develop a combined gradient–restoration algorithm in analogy with Section 5, a family of gradient–restoration

algorithms in analogy with Section 6, a hybrid modification of the sequential gradient-restoration algorithm in analogy with Section 7, and a sequential conjugate gradient-restoration algorithm in analogy with Section 8.

For the sake of brevity, all of the above extensions are omitted. For the construction of the sequential conjugate-gradient restoration algorithm for Problem (B-1), the reader is referred to Ref. 77 as well as to Refs. 40-41.

10. Multiple Subarcs

We complete our review by considering the following modification of Problem (B-1).

Problem (C-1). Minimize the functional

$$I = \int_0^3 f(x, u, \pi, t) dt + [g(x, \pi, t)]_3, \quad (164)$$

with respect to the state $x(t)$, the control $u(t)$, and the parameter π which satisfy the constraints

$$\dot{x} - \varphi(x, u, \pi, t) = 0, \quad 0 \leq t \leq 3, \quad (165)$$

$$S(x, u, \pi, t) = 0, \quad 0 \leq t \leq 3, \quad (166)$$

$$(x)_0 = \text{given}, \quad (167)$$

$$[M(x, \pi, t)]_1 = 0, \quad (168)$$

$$[E(x, \pi, t)]_2 = 0, \quad (169)$$

$$[\psi(x, \pi, t)]_3 = 0. \quad (170)$$

In Eqs. (164)–(170), the quantities I, f, g, t are scalar, and the quantities $x, u, \pi, \varphi, \psi, S, M, E$ are vectors having dimensions n, m, p, n, q, r, a, b , respectively.

It is assumed that the class of arcs being investigated includes three subarcs (an initial subarc, a middle subarc, and a terminal subarc) and that the analytical form of the functions f, φ, S might change from one subarc to another. Therefore

$$f(x, u, \pi, t) = \begin{cases} f_I(x, u, \pi, t), & 0 \leq t \leq 1, \\ f_{II}(x, u, \pi, t), & 1 \leq t \leq 2, \\ f_{III}(x, u, \pi, t), & 2 \leq t \leq 3, \end{cases} \quad (171)$$

and

$$\varphi(x, u, \pi, t) = \begin{cases} \varphi_I(x, u, \pi, t), & 0 \leq t \leq 1, \\ \varphi_{II}(x, u, \pi, t), & 1 \leq t \leq 2, \\ \varphi_{III}(x, u, \pi, t), & 2 \leq t \leq 3, \end{cases} \quad (172)$$

and

$$S(x, u, \pi, t) = \begin{cases} S_I(x, u, \pi, t), & 0 \leq t \leq 1, \\ S_{II}(x, u, \pi, t), & 1 \leq t \leq 2, \\ S_{III}(x, u, \pi, t), & 2 \leq t \leq 3. \end{cases} \quad (173)$$

Problem (C-1) differs from Problem (B-1) of Section 9 because of the presence of multiple subarcs and because the analytical form of the functions f , φ , S might change from one subarc to another. While in Problem (B-1) the control $u(t)$ and the multiplier $\lambda(t)$ are viewed to be continuous functions of the time, in the present Problem (C-1) control discontinuities and multiplier discontinuities are allowed at the points of junction between different subarcs.

The formulation represented by Problem (C-1) is particularly relevant to those problems of optimal control involving bounds on the control or the state or the time derivative of the state. For these problems, one might be unwilling to accept the simplistic view of a continuous extremal arc. Indeed, one might want to take the more realistic view of an extremal arc composed of several subarcs, some internal to the boundary being considered and some lying on the boundary (Refs. 69-70 and 74-77).

At any rate, basic to the point of view of this section is some advance information on the number and sequence of subarcs composing the extremal arc. This advance information can usually be obtained by means of a preliminary study of the continuous extremal arc model, made in accordance with Section 3 or Section 9.

Remark. The three-subarc formulation is used in Problem (C-1) to fix the ideas. By minor changes, one can deduce the formulas valid for a two-subarc formulation or generalize Problem (C-1) to an N -subarc formulation.

10.1. First-Order Conditions. The previous Problem (C-1) is one of the Bolza type and is characterized by the augmented functional⁹

$$J = \int_0^3 [f + \lambda^T(\dot{x} - \varphi) + \rho^T S] dt + (\sigma^T M)_1 + (\eta^T E)_2 + (g + \mu^T \psi)_3, \quad (174)$$

⁹ Since the integrand in (174) may be discontinuous at $t = 1$ and $t = 2$, the integral (174) must be computed as the sum of three integrals, one relative to the interval $0 \leq t \leq 1$, one relative to the interval $1 \leq t \leq 2$, and one relative to the interval $2 \leq t \leq 3$. An analogous remark holds for every other integral appearing in this section.

where $\lambda, \rho, \sigma, \eta, \mu$ are vector Lagrange multipliers having dimensions n, r, a, b, q , respectively. It is known that the minimization of (164), subject to (165)–(170), is identical with the minimization of (174), subject to (165)–(170), regardless of the choice of the multipliers $\lambda(t), \rho(t), \sigma, \eta, \mu$.

If the multipliers $\lambda(t), \rho(t), \sigma, \eta, \mu$ are chosen consistently with

$$\lambda - f_x + \varphi_x \lambda - S_x \rho = 0, \quad 0 \leq t \leq 3, \quad (175)$$

$$(\lambda_+ - \lambda_- - M_x \sigma)_1 = 0, \quad (176)$$

$$(\lambda_+ - \lambda_- - E_x \eta)_2 = 0, \quad (177)$$

$$(\lambda + g_x + \psi_x \mu)_3 = 0, \quad (178)$$

then the optimum solution $x(t), u(t), \pi$ must satisfy the following relations:

$$f_u - \varphi_u \lambda + S_u \rho = 0, \quad 0 \leq t \leq 3, \quad (179)$$

$$\int_0^3 (f_\pi - \varphi_\pi \lambda + S_\pi \rho) dt + (M_\pi \sigma)_1 + (E_\pi \eta)_2 + (g_\pi + \psi_\pi \mu)_3 = 0. \quad (180)$$

The system composed of the feasibility equations (165)–(170) and the optimality conditions (175)–(180) constitutes a nonlinear, multi-point boundary-value problem in which the unknowns are the functions $x(t), u(t), \pi$ and the multipliers $\lambda(t), \rho(t), \sigma, \eta, \mu$. Rather than solving this system directly, we shall solve it indirectly by means of a sequential gradient-restoration algorithm, with complete restoration (SGRA-CR), similar in construction to that of Section 3. Prior to doing so, we shall define convergence in a numerical sense through Ineqs. (16), where P and Q are now the scalar performance indexes associated with Problem (C-1). These performance indexes are given by

$$P = \int_0^3 N(\dot{x} - \varphi) dt + \int_0^3 N(S) dt + N(M)_1 + N(E)_2 + N(\psi)_3, \quad (181)$$

$$\begin{aligned} Q = & \int_0^3 N(\lambda - f_x + \varphi_x \lambda - S_x \rho) dt + \int_0^3 N(f_u - \varphi_u \lambda + S_u \rho) dt \\ & + N \left[\int_0^3 (f_\pi - \varphi_\pi \lambda + S_\pi \rho) dt + (M_\pi \sigma)_1 + (E_\pi \eta)_2 + (g_\pi + \psi_\pi \mu)_3 \right] \\ & + N(\lambda_+ - \lambda_- - M_x \sigma)_1 + N(\lambda_+ - \lambda_- - E_x \eta)_2 + N(\lambda + g_x + \psi_x \mu)_3 \end{aligned} \quad (182)$$

and measure the errors in the constraints and the optimality conditions, respectively.

10.2. Sequential Gradient-Restoration Algorithm. In constructing a sequential gradient-restoration algorithm for Problem (C-1), we shall retain the properties of Section 3.1 and the notation of Section 3.2. We shall represent the varied functions $\tilde{x}(t)$, $\tilde{u}(t)$, $\tilde{\pi}$ through Eqs. (23). Therefore, we are faced with two problems: (a) the determination of the basic functions $A(t)$, $B(t)$, C and (b) the determination of the stepsize α .

10.3. Determination of the Basic Functions. The determination of the basic functions $A(t)$, $B(t)$, C is accomplished through the formulation of the following auxiliary minimization problems.

Problem (C-2). For the gradient phase, minimize the functional

$$\delta I = \int_0^3 \delta f \, dt + (\delta g)_3, \quad (183)$$

with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 3, \quad (184)$$

$$\delta(S) = 0, \quad 0 \leq t \leq 3, \quad (185)$$

$$\delta(x)_0 = 0, \quad (186)$$

$$\delta(M)_1 = 0, \quad (187)$$

$$\delta(E)_2 = 0, \quad (188)$$

$$\delta(\psi)_3 = 0, \quad (189)$$

and the isoperimetric constraint

$$K = \int_0^3 \Delta u^T \Delta u \, dt + \Delta \pi^T \Delta \pi. \quad (190)$$

Problem (C-3). For the restoration phase, minimize the functional (190), with respect to the state change $\Delta x(t)$, the control change $\Delta u(t)$, and the parameter change $\Delta \pi$ which satisfy the linearized constraints

$$\delta(\dot{x} - \varphi) + \alpha(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 3, \quad (191)$$

$$\delta(S) + \alpha S = 0, \quad 0 \leq t \leq 3, \quad (192)$$

$$\delta(x)_0 = 0, \quad (193)$$

$$\delta(M)_1 + \alpha(M)_1 = 0, \quad (194)$$

$$\delta(E)_2 + \alpha(E)_2 = 0, \quad (195)$$

$$\delta(\psi)_3 + \alpha(\psi)_3 = 0. \quad (196)$$

10.4. Linear, Multi-Point Boundary-Value Problem. Problems (C-2) and (C-3) are variational problems of the Bolza type. The form of the linearized constraints and optimality conditions is such that the equations governing Problems (C-2) and (C-3) can be embedded in the following single system:

$$\dot{A} - \varphi_x^T A - \varphi_u^T B - \varphi_\pi^T C + k_r(\dot{x} - \varphi) = 0, \quad 0 \leq t \leq 3, \quad (197)$$

$$S_x^T A + S_u^T B + S_\pi^T C + k_r S = 0, \quad 0 \leq t \leq 3, \quad (198)$$

$$(A)_0 = 0, \quad (199)$$

$$(M_x^T A + M_\pi^T C + k_r M)_1 = 0, \quad (200)$$

$$(E_x^T A + E_\pi^T C + k_r E)_2 = 0, \quad (201)$$

$$(\psi_x^T A + \psi_\pi^T C + k_r \psi)_3 = 0, \quad (202)$$

and

$$\dot{\lambda} - k_g f_x + \varphi_x \lambda - S_x \rho = 0, \quad 0 \leq t \leq 3, \quad (203)$$

$$(\lambda_+ - \lambda_- - M_x \sigma)_1 = 0, \quad (204)$$

$$(\lambda_+ - \lambda_- - E_x \eta)_2 = 0, \quad (205)$$

$$(\lambda + k_g g_x + \psi_x \mu)_3 = 0, \quad (206)$$

$$B + k_g f_u - \varphi_u \lambda + S_u \rho = 0, \quad 0 \leq t \leq 3, \quad (207)$$

$$C + \int_0^3 (k_g f_\pi - \varphi_\pi \lambda + S_\pi \rho) dt + (M_\pi \sigma)_1 + (E_\pi \eta)_2 + (k_g g_\pi + \psi_\pi \mu)_3 = 0. \quad (208)$$

The constants k_g and k_r appearing in (197)–(208) take the values (41)–(42).

For given nominal functions $x(t)$, $u(t)$, π and constants k_g and k_r , Eqs. (197)–(208) define the functions $A(t)$, $B(t)$, C and the multipliers $\lambda(t)$, $\rho(t)$, σ , η , μ . Clearly, Eqs. (197)–(208) constitute a linear, multi-point boundary-value problem, which can be solved independently of the value assigned to the stepsize α .

10.5. Method of Particular Solutions. The usual approach of backward integration of the λ -equation and forward integration of the A -equation (see Section 3.5) cannot be employed with this class of problems. This is due to the fact that the computation of the multiplier $\rho(t)$ over the interval $0 \leq t \leq 3$ requires the simultaneous integration of both the λ -equation and the A -equation. Because of the coupling, the total system must be integrated simultaneously in either the forward direction or the backward direction. Since either direction requires the same number of integrations, we choose the forward direction.

Let a *sweep* be defined as a forward integration of Eqs. (197)–(208), obtained by specifying the $n + p + a + b$ components of the vector

$$z = [\lambda^T(0), C^T, \sigma^T, \eta^T]^T \quad (209)$$

and satisfying all of the equations, except (200), (201), (202), (206), and (208). Note that the successful execution of the integration is subordinated to the condition that

$$\det[S_u^T S_u] \neq 0, \quad 0 \leq t \leq 3. \quad (210)$$

This condition arises by observing that, at each time station t , and for given values of A , λ , and C , relations (198) and (207) constitute a system of $m + r$ equations in the $m + r$ components of the vectors B and ρ . This system admits a unique solution providing (210) is satisfied.

If the method of particular solutions is employed, one must execute $n + p + a + b + 1$ independent sweeps. Specifically, the first $n + p + a + b$ sweeps are performed by considering the identity matrix of order $n + p + a + b$ and by choosing the vector z associated with the i th integration, $i = 1, \dots, n + p + a + b$, as identical with the i th column of the above identity matrix. The last sweep is performed by choosing the vector z associated with the $(n + p + a + b + 1)$ th integration as identical with the null vector.

Next, we combine the particular solutions linearly and write the composite solution in the form

$$A(t) = \sum k_i A_i(t), \quad B(t) = \sum k_i B_i(t), \quad C = \sum k_i C_i, \quad (211)$$

$$\lambda(t) = \sum k_i \lambda_i(t), \quad \rho(t) = \sum k_i \rho_i(t), \quad \sigma = \sum k_i \sigma_i, \quad \eta = \sum k_i \eta_i, \quad (212)$$

where $i = 1, \dots, n + p + a + b + 1$. The $n + p + a + b + 1$ constants k_i are determined simultaneously with the q components of the multiplier μ by satisfying the linear system of order $n + p + q + a + b + 1$ composed of the n equations (206), the p equations (208), the q equations (202), the a equations (200), the b equations (201), and the supplementary scalar relation $\sum k_i = 1$, which arises from the nonhomogeneity of the system (197)–(208).

10.6. Computational Considerations. Each sweep involves integrating $2n$ differential equations, that is, the n linearized state equations (197) and the n multiplier equations (203). Since $n + p + a + b + 1$ sweeps are involved, the amount of computational work performed per iteration, gradient or restorative, is proportional to the factor

$$w = 2n(n + p + a + b + 1), \quad (213)$$

regardless of the number of final conditions q . Clearly, the computational effort per iteration increases with the square of the dimension of the state vector. Therefore, in devising transformations leading to problems of the form (164)–(170), one must keep in mind the following: whenever possible, it is a good idea to prevent or limit an increase in the dimension of the state vector x .

10.7. Determination of the Stepsize. The functions $A(t)$, $B(t)$, C solving Eqs. (197)–(208) are such that the first variation properties (48)–(49) hold. This being the case, it is possible to determine the stepsize α so that the descent property (24-1) is enforced for the gradient phase and the descent property (24-2) is enforced for the restoration phase.

Specifically, one has to consider the one-parameter family of solutions (50) for which the augmented functional (174) and the constraint error (181) take the form (51). Then, the stepsize α must be determined so that Ineqs. (52) are enforced for each gradient iteration and Ineq. (53) is enforced for each restorative iteration. To this effect, one can employ the search techniques outlined in Sections 3.9 and 3.10.

An order-of-magnitude analysis shows that (60)–(61) hold for the present problem. Hence, by a suitable choice of the gradient stepsize, the fundamental stability property (59) can be enforced for each complete gradient–restoration cycle.

10.8. Generalizations and Extensions. For Problem (C-1), generalizations and extensions similar to those shown for Problem (A-1) are possible. Specifically, one can develop a combined gradient–restoration algorithm in analogy with Section 5, a family of gradient–restoration algorithms in analogy with Section 6, a hybrid modification of the sequential gradient–restoration algorithm in analogy with Section 7, and a sequential conjugate gradient–restoration algorithm in analogy with Section 8. For the sake of brevity, all of the above generalizations and extensions are omitted.

11. Appendix: Transformation Techniques

In the previous sections, we described gradient–restoration algorithms in connection with three principal problems of optimal control, namely, Problems (A-1), (B-1), and (C-1). Problem (A-1), represented by (1)–(4), is concerned with the minimization of a functional subject to differential constraints, initial conditions, and final conditions.

Problem (B-1), represented by (129)–(133), is the same as Problem (A-1) complicated by the presence of nondifferential constraints. And Problem (C-1), represented by (164)–(170), is the same as Problem (B-1) complicated by the presence of multiple subarcs and different specifications of the functions under consideration along the multiple subarcs.

In this section, we describe some transformation techniques useful to bring a wide variety of problems of optimal control within the framework of the models represented by Problems (A-1), (B-1), and (C-1). More specifically, the following topics are treated in general here: (i) time normalization; (ii) free initial state; (iii) bounded control; and (iv) bounded state. In actual practice, the transformations pertaining to (ii), (iii), (iv) usually precede the transformation pertaining to (i). For illustrative examples, the reader is referred to Refs. 12–43 as well as to Refs. 60–77.

11.1. Time Normalization. Consider an optimal control problem characterized by the fact that the running time θ is not normalized. Denote by τ the actual final time, that is, suppose that

$$0 \leq \theta \leq \tau. \quad (214)$$

We wish to replace the running time θ with the normalized time t , defined in such a way that, associated with each subarc composing the extremal arc, is a normalized time length

$$\Delta t = 1. \quad (215)$$

One Subarc. Assume that the extremal arc 01 includes a single subarc. Denote by $\theta_0 = 0$ the initial time and by $\theta_1 = \tau$ the final time. The conversion to problems where the time t is normalized is done through the transformation (Refs. 63 and 88)

$$\theta = \tau t, \quad 0 \leq t \leq 1. \quad (216)$$

The quantity τ , if it is free, is regarded to be an additional component of the vector parameter π being optimized.

Two Subarcs. Next, assume that the extremal arc 012 includes two subarcs, an initial subarc 01 and a final subarc 12. Denote by $\theta_0 = 0$ the initial time, by θ_1 the time of exit from the first subarc, and by $\theta_2 = \tau$ the final time. The conversion to problems where the time t is normalized is done through the transformation

$$\theta = \theta_1 t, \quad 0 \leq t \leq 1, \quad (217-1)$$

$$\theta = \theta_1 + (\tau - \theta_1)(t - 1), \quad 1 \leq t \leq 2. \quad (217-2)$$

The quantities θ_1 and τ , if they are free, are regarded to be additional components of the vector parameter π being optimized.

Three Subarcs. Finally, assume that the extremal arc 0123 includes three subarcs, an initial subarc 01, a middle subarc 12, and a final subarc 23. Denote by $\theta_0 = 0$ the initial time, by θ_1 the time of exit from the first subarc, by θ_2 the time of exit from the second subarc, and by $\theta_3 = \tau$ the final time. The conversion to problems where the time t is normalized is done through the transformation (Refs. 69 and 74-77)

$$\theta = \theta_1 t, \quad 0 \leq t \leq 1, \quad (218-1)$$

$$\theta = \theta_1 + (\theta_2 - \theta_1)(t - 1), \quad 1 \leq t \leq 2, \quad (218-2)$$

$$\theta = \theta_2 + (\tau - \theta_2)(t - 2), \quad 2 \leq t \leq 3. \quad (218-3)$$

The quantities θ_1 , θ_2 , τ , if they are free, are regarded to be additional components of the vector parameter π being optimized.

Examples of optimal control problems treated by means of the above transformations are given in Refs. 63, 69, 71, and 74-77.

11.2. Free Initial State. Suppose that the k th component of the state vector $x(\theta)$, say $y(\theta)$, is free at the initial point. That is, assume that $y(0)$ is to be determined from the solution of the optimal control problem.

The conversion to problems with fixed initial state can be done by employing one of the following transformations:

$$z(\theta) = y(\theta) - y(0), \quad 0 \leq \theta \leq \tau, \quad (219-1)$$

$$z(0) = 0, \quad (219-2)$$

or

$$z(\theta) = y(\theta)/y(0), \quad 0 \leq \theta \leq \tau, \quad (220-1)$$

$$z(0) = 1, \quad (220-2)$$

where $z(\theta)$ denotes an additional state variable. Clearly, transformation (219) uses the state difference and transformation (220) uses the state ratio.

Next, a substitution technique is employed, the state component $y(\theta)$ is eliminated from the problem using (219) or (220), and the state vector $x(\theta)$ is redefined so that $z(\theta)$ replaces $y(\theta)$ as the k th component of $x(\theta)$. With this understanding, we are now faced with an optimal control problem involving a fixed initial state. In this formulation, the free quantity $y(0)$ becomes an additional component of the vector parameter π being optimized.

Transformation (219) is to be preferred to transformation (220), because it avoids increasing the nonlinearity of the optimal control problem. In particular, if the original optimal control problem is linear-quadratic, transformation (219) leads to another linear-quadratic problem, while this is not the case with transformation (220). Also, transformation (219) avoids the complications which would or could arise with transformation (220), should the optimal solution be such that

$$y(0) = 0 \quad \text{or} \quad y(0) \cong 0. \quad (221)$$

Examples of optimal control problems treated by means of the above transformations are given in Refs. 42-43.

11.3. Bounded Control. One-Sided Inequality. Suppose that the k th component of the control vector $u(\theta)$, say $v(\theta)$, is subject to a one-sided inequality of the form

$$v \geq a, \quad 0 \leq \theta \leq \tau, \quad (222)$$

where a denotes a lower bound on the control. This lower bound can have the form

$$a = \text{const}, \quad \text{or} \quad a = a(\theta), \quad \text{or} \quad a = a(x, \pi, \theta). \quad (223)$$

There are two main approaches that can be employed in solving optimal control problems of this type. One is the indirect approach, in which the extremal arc is viewed as a single subarc, even though a portion of it may lie extremely close to the control boundary. The other is the direct approach, in which the extremal arc is viewed as being composed of several subarcs, some internal to the control boundary and some lying on the control boundary.

Indirect Approach. The conversion to problems of the type considered in this paper can be done by employing the transformation

$$v = a + w^2, \quad 0 \leq \theta \leq \tau, \quad (224)$$

where $w(\theta)$ denotes the auxiliary control variable.

Substitution Technique. In this technique, the control component $v(\theta)$ is eliminated from the problem using (224), and the control vector $u(\theta)$ is redefined so that $w(\theta)$ replaces $v(\theta)$ as the k th component of $u(\theta)$. With this understanding, we are now faced with an optimal control problem involving an unbounded control.

Nondifferential Constraint Technique. In this technique, both $v(\theta)$ and $w(\theta)$ are retained in the formulation of the problem; that is, the control vector $u(\theta)$ is redefined so as to include $w(\theta)$ as an additional component. However, it should be understood that the components of this augmented control are not free, but must be chosen consistently with (224), which becomes a nondifferential constraint to be satisfied everywhere along the interval of integration.

Examples of optimal control problems treated by means of the above transformation are given in Ref. 32.

Direct Approach. Suppose that the optimal control problem is first solved ignoring Ineq. (222). Further, suppose that the ensuing solution violates Ineq. (222) over the terminal portion of the trajectory. Then, it is natural to postulate an extremal arc including two subarcs, specifically,

$$w \neq 0, \quad 0 \leq \theta \leq \theta_1, \quad (225-1)$$

$$w = 0, \quad \theta_1 \leq \theta \leq \tau, \quad (225-2)$$

with the implication that

$$v = a + w^2, \quad 0 \leq \theta \leq \theta_1, \quad (226-1)$$

$$v = a, \quad \theta_1 \leq \theta \leq \tau. \quad (226-2)$$

Substitution Technique. In this technique, the control component $v(\theta)$ is eliminated from the problem using (226), and the control vector $u(\theta)$ is redefined so that $w(\theta)$ replaces $v(\theta)$ as the k th component of $u(\theta)$.

Nondifferential Constraint Technique. In this technique, both $v(\theta)$ and $w(\theta)$ are retained in the formulation of the problem; that is, the control vector $u(\theta)$ is redefined so as to include $w(\theta)$ as an additional component. However, it should be understood that the components of this augmented control are not free but must be chosen consistently with (226).

Remark. The generalization of the nondifferential constraint technique to the case of a control inequality constraint of the form

$$L(x, u, \pi, \theta) \geq 0, \quad 0 \leq \theta \leq \tau, \quad (227)$$

is immediate in both the direct approach and the indirect approach. Therefore, this generalization is omitted here, for the sake of brevity.

11.4. Bounded Control. Two-Sided Inequality. Suppose that a particular component of the control vector $u(\theta)$, say $v(\theta)$, is subject to a two-sided inequality constraint of the form

$$a \leq v \leq b, \quad 0 \leq \theta \leq \tau, \quad (228)$$

where a and b denote lower and upper bounds, respectively. These bounds can have the form

$$a = \text{const}, \quad \text{or} \quad a = a(\theta), \quad \text{or} \quad a = a(x, \pi, \theta), \quad (229)$$

$$b = \text{const}, \quad \text{or} \quad b = b(\theta), \quad \text{or} \quad b = b(x, \pi, \theta). \quad (230)$$

The conversion to problems of the type considered in this paper can be done by employing one of the following transformations:

$$v = a + (b - a) \sin^2 w, \quad 0 \leq \theta \leq \tau, \quad (231)$$

or

$$v = \frac{1}{2}(b + a) + \frac{1}{2}(b - a) \sin w, \quad 0 \leq \theta \leq \tau, \quad (232)$$

where $w(\theta)$ denotes an auxiliary control variable. Alternatively, if one wishes to avoid the computation of trigonometric functions, the nondifferential constraint (231) or (232) can be replaced by

$$(v - a)(b - v) - w^2 = 0, \quad 0 \leq \theta \leq \tau. \quad (233)$$

Transformations (231), (232), or (233) can be employed in analogy with transformation (224) of Section 11.3 within the framework of either an indirect approach or a direct approach and with reference to either a substitution technique or a nondifferential constraint technique.¹⁰ Because of the analogy, the detailed treatment is omitted, for the sake of brevity.

Examples of optimal control problems treated by means of the above transformations are given in Ref. 32.

Remark. The generalization of the nondifferential constraint technique to the case of a control inequality constraint of the form

$$a \leq L(x, u, \pi, \theta) \leq b, \quad 0 \leq \theta \leq \tau, \quad (234)$$

is immediate in both the direct approach and the indirect approach. Therefore, this generalization is omitted here, for the sake of brevity.

¹⁰ However, use of transformation (233) is recommended only in connection with a nondifferential constraint technique.

11.5. Bounded State. Suppose that the state vector x is subject to a one-sided inequality constraint of the form

$$L(x, \theta) \geq 0, \quad 0 \leq \theta \leq \tau. \quad (235)$$

Assume that the state inequality constraint (235) is of order k . This means that the k th time derivative of the function $L(x, \theta)$ is the first to contain the control u explicitly.¹¹ For simplicity, assume that the parameter π is not present in the original formulation of the optimal control problem (that is, the formulation in which the nonnormalized running time θ is employed as the independent variable).

There are two main approaches which can be employed in solving optimal control problems of this type. One is the indirect approach, in which the extremal arc is viewed as a single subarc, even though a portion of it may lie extremely close to the state boundary. The other is the direct approach, in which the extremal arc is viewed as being composed of several subarcs, some internal to the state boundary and some lying on the state boundary.

Indirect Approach. The conversion to problems of the type considered in this paper is done as follows. First, we introduce the auxiliary state vector $y(\theta)$ and the auxiliary control variable $w(\theta)$. We require the k -vector y and the scalar w to satisfy the differential equations

$$dy_i/d\theta = y_{i+1}, \quad i = 1, \dots, k-1, \quad 0 \leq \theta \leq \tau, \quad (236)$$

$$dy_k/d\theta = w, \quad 0 \leq \theta \leq \tau. \quad (237)$$

Next, we discard the state inequality constraint (235) and replace it with the state equality constraint

$$L(x, \theta) = y_1^2, \quad 0 \leq \theta \leq \tau, \quad (238)$$

where $y_1(\theta)$ denotes the first component of the auxiliary state vector. Then, we differentiate Eq. (238) k times with respect to θ and obtain the following relations:

$$L_i(x, \theta) - Q_i(y) - 2y_1 y_{i+1} = 0, \quad i = 1, \dots, k-1, \quad 0 \leq \theta \leq \tau, \quad (239)$$

$$L_k(x, u, \theta) - Q_k(y) - 2y_1 w = 0, \quad 0 \leq \theta \leq \tau. \quad (240)$$

Here, the symbols $L_i(x, \theta)$ and $L_k(x, u, \theta)$ denote the successive total derivatives of $L(x, \theta)$, and the symbols $Q_i(y)$ denote known quadratic functions of the auxiliary state vector y , specifically,

$$Q_1(y) = 0, \quad Q_2(y) = 2y_2^2, \quad Q_3(y) = 6y_2 y_3, \quad Q_4(y) = 6y_3^2 + 8y_2 y_4, \dots \quad (241)$$

¹¹ In actual practice, the most common types of state inequality constraints have order $k = 1$ or $k = 2$.

We note that (238)–(239) are successive first integrals of (240). Also, (238)–(239), when applied at the initial point, yield the initial condition for the auxiliary state vector y . Since $x(0)$ is given, (238)–(239) imply that

$$y(0) = \text{given.} \quad (242)$$

Consequently, we conclude that the state inequality constraint (235) can be replaced by the state equality constraint (238), which in turn can be replaced by the control equality constraint (240) in combination with the differential equations (236)–(237) and the initial condition (242).

When this approach is employed, the state vector $x(\theta)$ must be redefined so as to include $y_1(\theta), y_2(\theta), \dots, y_k(\theta)$ as additional components. Analogously, the control vector $u(\theta)$ must be redefined so as to include $w(\theta)$ as an additional component.¹² Clearly, then, the dimension of the state vector is increased by k (the order of the state inequality constraint), and the dimension of the control vector is increased by one.

Examples of optimal control problems treated by means of the above transformation are given in Refs. 31–33 and 71 as well as in Ref. 123.

Direct Approach. This approach is now explained in connection with several transformation techniques: a standard transformation technique, a nonstandard transformation technique, and a hybrid transformation technique.

Standard Transformation Technique. Suppose that the optimal control problem is first solved ignoring Ineq. (235). Further, suppose that the ensuing solution violates Ineq. (235) over both the initial portion of the trajectory and the terminal portion of the trajectory. Then, it is natural to postulate an extremal arc including three subarcs, specifically,

$$L(x, \theta) \geq 0, \quad 0 \leq \theta \leq \theta_1, \quad (243-1)$$

$$L(x, \theta) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (243-2)$$

$$L(x, \theta) \geq 0, \quad \theta_2 \leq \theta \leq \tau. \quad (243-3)$$

Equations (236)–(238) and (243) imply that

$$w \neq 0, \quad 0 \leq \theta \leq \theta_1, \quad (244-1)$$

$$w = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (244-2)$$

$$w \neq 0, \quad \theta_2 \leq \theta \leq \tau. \quad (244-3)$$

¹² Note that the components of the augmented control are not free, but must be chosen consistently with (240), which becomes a nondifferential constraint to be satisfied everywhere along the interval of integration.

Therefore, Eqs. (236), (237), (240), and (242) must be employed in conjunction with the following provisions, pertaining to (237) and (240):

$$dy_k/d\theta = w, \quad 0 \leq \theta \leq \theta_1, \quad (245-1)$$

$$dy_k/d\theta = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (245-2)$$

$$dy_k/d\theta = w, \quad \theta_2 \leq \theta \leq \tau, \quad (245-3)$$

and

$$L_k(x, u, \theta) - Q_k(y) - 2y_1w = 0, \quad 0 \leq \theta \leq \theta_1, \quad (246-1)$$

$$L_k(x, u, \theta) - Q_k(y) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (246-2)$$

$$L_k(x, u, \theta) - Q_k(y) - 2y_1w = 0, \quad \theta_2 \leq \theta \leq \tau. \quad (246-3)$$

In addition, the following entrance condition to the state boundary must be considered:

$$y(\theta_1) = 0. \quad (247)$$

When the standard transformation technique is employed, the state vector $x(\theta)$ must be redefined so as to include $y(\theta)$. Analogously, the control vector $u(\theta)$ must be redefined so as to include $w(\theta)$. Therefore, the dimension of the state vector is increased by k and the dimension of the control vector is increased by one.

Examples of optimal control problems treated by means of the above transformation are given in Refs. 34-35 and 74.

Nonstandard Transformation Technique. There are many problems of optimal control in which the function $L(x, \theta)$ is such that its k th derivative $L_k(x, u, \theta)$ has a constant sign along each subarc composing the extremal arc. Whenever this situation arises, substantial simplifications are possible.

To fix the ideas, consider a three-subarc situation, in accordance with (243). The assumption of piecewise monotonicity of $L_k(x, u, \theta)$ is consistent with (243) and with the entrance conditions

$$[L(x, \theta)]_1 = 0 \quad \text{and} \quad [L_i(x, \theta)]_1 = 0, \quad i = 1, \dots, k-1, \quad (248)$$

providing the function $L_k(x, u, \theta)$ behaves as follows:

$$L_k(x, u, \theta) \leq 0, \quad 0 \leq \theta \leq \theta_1, \quad (249-1)$$

$$L_k(x, u, \theta) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (249-2)$$

$$L_k(x, u, \theta) \geq 0, \quad \theta_2 \leq \theta \leq \tau, \quad (249-3)$$

or

$$L_k(x, u, \theta) \geq 0, \quad 0 \leq \theta \leq \theta_1, \quad (250-1)$$

$$L_k(x, u, \theta) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (250-2)$$

$$L_k(x, u, \theta) \geq 0, \quad \theta_2 \leq \theta \leq \tau, \quad (250-3)$$

where (249) apply if k is odd and (250) apply if k is even. In order to ensure satisfaction of conditions (249) or (250) over the interval $0 \leq \theta \leq \tau$, we introduce the auxiliary control variable $w(\theta)$ such that

$$L_k(x, u, \theta) + w^2 = 0, \quad 0 \leq \theta \leq \theta_1, \quad (251-1)$$

$$L_k(x, u, \theta) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (251-2)$$

$$L_k(x, u, \theta) - w^2 = 0, \quad \theta_2 \leq \theta \leq \tau, \quad (251-3)$$

or

$$L_k(x, u, \theta) - w^2 = 0, \quad 0 \leq \theta \leq \theta_1, \quad (252-1)$$

$$L_k(x, u, \theta) = 0, \quad \theta_1 \leq \theta \leq \theta_2, \quad (252-2)$$

$$L_k(x, u, \theta) - w^2 = 0, \quad \theta_2 \leq \theta \leq \tau, \quad (252-3)$$

where (251) apply if k is odd and (252) apply if k is even.

When the nonstandard transformation technique is employed, the state vector $x(\theta)$ remains unchanged, while the control vector $u(\theta)$ is redefined so as to include $w(\theta)$ as an additional component. Therefore, the dimension of the control vector is increased by one.

Examples of successful application of the nonstandard transformation technique are given in Refs. 38–39 and 75–76.

Hybrid Transformation Technique. An alternative to the standard transformation technique is the hybrid transformation technique developed in Refs. 27–28 and 69. In this hybrid transformation technique, the state vector $x(\theta)$ is augmented so as to include the components $z_1(\theta), z_2(\theta), \dots, z_h(\theta)$ of an auxiliary state vector $z(\theta)$, with

$$0 \leq h \leq k. \quad (253)$$

The number h of auxiliary state variables is equal to the number of nonlinear components of the entrance conditions (248). After the augmented state vector is introduced, the entrance conditions (248) can be rewritten in linear form. As a consequence, it becomes possible to exploit the anchoring properties established in Ref. 70. These anchoring properties, if employed in connection with the sequential gradient–

restoration algorithm, enable one to satisfy the state inequality constraint (235) everywhere at the end of each gradient iteration and each restorative iteration. Therefore, satisfaction of the state inequality constraint at convergence is guaranteed. The details of this hybrid transformation technique are omitted for the sake of brevity, and the reader is referred to Ref. 69.

Examples of successful application of the hybrid transformation technique are given in Refs. 27–28 and 69.

Remark. Whenever possible, the nonstandard transformation technique should be preferred to both the standard transformation technique and the hybrid transformation technique. This is due to the fact that the dimension of the state vector remains unchanged in the nonstandard transformation, while it increases by k in the standard transformation and by h in the hybrid transformation. Indeed, as noted in Section 10, the computational effort per iteration increases with the square of the dimension of the state vector.

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